ANNUAL SURVEY OF MICRO, SMALL AND MEDIUM ENTERPRISES IN INDIA

THE ROLE OF DIGITALISATION IN ENTERPRISE DEVELOPMENT

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MARCH 2025

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Annual Survey of Micro, Small and Medium Enterprises (MSMEs) in India:
The Role of Digitalisation in Enterprise Development
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ISBN: 978-81-986536-3-5

Published by the Indian Council for Research on International Economic Relations (ICRIER) in March 2025

Recommended citation: Goyal, T.M., Puri, H. & Khanna, R. (2025). Annual Survey of Micro, Small and Medium Enterprises (MSMEs) in India: The Role of Digitalisation in Enterprise Development. Indian Council for Research on International Economic Relations (ICRIER)

TABLE OF CONTENTS

PREFACE

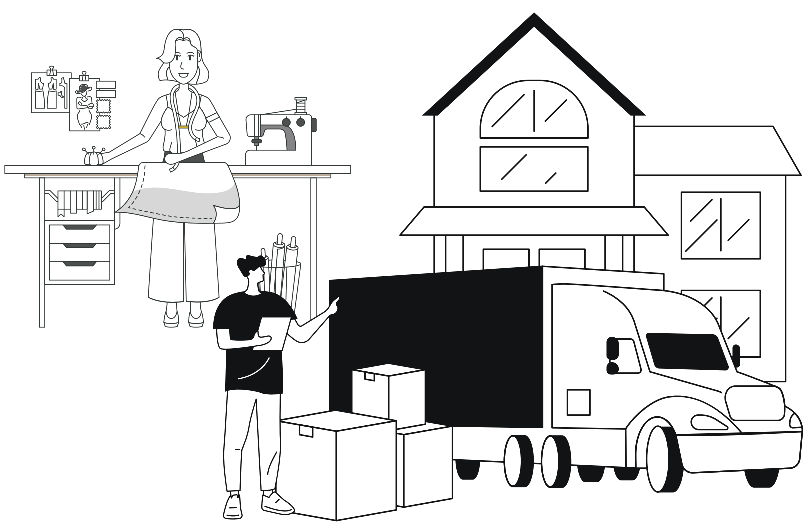
This is the third edition of ICRIER's annual survey on micro, small, and medium enterprises (MSMEs) in India. The survey aims to assess the impact of digitalisation, particularly e-commerce platforms, on MSMEs' business growth and development. Using primary survey data, it focuses on understanding how digital adoption influences enterprise expansion and growth prospects in India.

ICRIER's MSME surveys were initiated in 2021 in response to the COVID-19 pandemic, which severely impacted the MSME sector. Identifying strategies that could help businesses survive during periods of physical restrictions became crucial. Digitalisation emerged as a key solution, as many firms turned to online platforms to reach markets. Consequently, in 2021, a survey of around 1,500 manufacturing MSMEs, integrated with e-commerce platforms was conducted. An inter-temporal assessment compared firms' post-pandemic performance to pre-pandemic levels, focusing on online sales to evaluate digitalisation's role in the survival of firms.

In the second edition (2022-23), the survey expanded into a longitudinal study with a cross-sectional sample, equally divided between e-commerce-integrated and non-integrated firms. This comparison aimed to evaluate how integration impacts business outcomes. Around 2,000 MSMEs were surveyed using purposive sampling and a cluster-based approach to minimise bias. Besides core business performance indicators such as sales, employment, and profit, the study also briefly assessed issues like MSMEs' access to finance.

For this edition, 2,365 firms were surveyed in 2024 to further compare integrated and non-integrated businesses. While previous surveys have established the key benefits, such as increased sales, higher profit margins, and positive effects on employment, this study also focuses on examining the second-order effects of digitalisation on MSMEs. Specifically, it explores how integration enhances access to finance, international markets (exports), and information—factors that play a crucial role in enterprise growth and development. Importantly, this year's survey design enables a time series and cross-sectional data analysis.

The report presents findings based on three main aspects: (a) business performance metrics, including sales, profitability, hiring, and research and development (R&D); (b) access to finance and related challenges; and (c) international market access. These insights help determine whether e-commerce-integrated firms have a higher potential for scaling and enterprise development than non-integrated firms.



ACKNOWLEDGEMENTS

The authors are grateful to the 2,365 enterprises that participated in the field survey across India, and to other stakeholders who participated through in-depth interviews. This study could not have been completed without their responses.

The authors would like to express their sincere gratitude to ICRIER's Director and Chief Executive Deepak Mishra for his continuous guidance and encouragement. His inputs have been extremely useful in lending a meaningful perspective to the findings of the survey.

The authors are also thankful to the survey partner Ipsos, particularly Gajanand Periwal and Mitali Singh, for conducting the survey and compiling the results.

During this study, the authors interacted with various stakeholders, including policymakers, e-commerce companies, industry associations, representatives from the industry, and academic experts. The authors thank them for sharing their valuable insights, which helped strengthen the analysis. The authors would like to extend their gratitude to Walmart for supporting this timely study.

Several experts were consulted while drafting the questionnaire, and their inputs were incorporated. The authors would particularly like to thank Tirumala KV and Renu Lata from the Ministry of Commerce and Industry, and Anil Bhardwaj from Federation of Indian Small and Medium Enterprise for their feedback and inputs. The authors are also grateful to BG Mahesh and his team at Sahamati for sharing data and insights on the Account Aggregator framework. The authors are thankful to Radhicka Kapoor and Shravani Prakash for their comments and feedback. Gratitude extends to Puja Mehra for her detailed review and editorial suggestions, which significantly improved the quality of the work.

The authors are also thankful to Harleen Kaur's finance and operations team, Neha Saini's administrative team, documentation officer Chhaya Singh, Information Technology manager Rajkumar Shahi, and Anvi Mehta's communications team for working tirelessly to ensure the research work was uninterrupted and is disseminated in an effective and timely manner.

The authors are also thankful to the Seema Sethi Design team for their time and effort towards designing this report, Preeti Lourdes John for copy editing and proof reading the report and to Genesis Printers for printing.

All errors are that of the authors.

LIST OF ABBREVIATIONS

AA	Account Aggregator
ADB	Asian Development Bank
ADBI	Asian Development Bank Institute
ASUSE	Annual Survey of Unincorporated Sector Enterprises
B20	Business 20
B2C	Business-to-Consumer
DPI	Digital Public Infrastructure
GVA	Gross Value Added
HSC	Higher Secondary Certificate
ICRIER	Indian Council for Research on International Economic Relations
INR	Indian Rupee
ITC	International Trade Centre
MFI	Microfinance Institution
MSE-CDP	Micro & Small Enterprises Cluster Development
MSMEs	Micro, Small and Medium Enterprises
MUDRA	Micro Units Development and Refinance Agency
NBFCs	Non-Banking Financial Corporations
NCR	National Capital Region
OECD	Organization for Economic Cooperation and Development
RBI	Reserve Bank of India
R&D	Research and Development
SME	Small and Medium Enterprise
SSC	Secondary School Certificate
UBS-ED	University of Stellenbosch Business School
ULI	Unified Lending Interface
UPI	Unified Payments Interface

Key Highlights of the Report

- **Post COVID-19, Indian MSMEs have demonstrated remarkable resilience.** Contrary to the common perception that MSMEs are generally vulnerable and distressed units, ICRIER's study, based on a survey of 2,365 Udyam-registered manufacturing sector MSMEs, indicates a robust performance in recent years. Since 2021, a majority of surveyed firms have seen sales growth, profit margins have improved, and hiring has increased—showcasing enterprise growth and development.

- **Digitalisation is a key driver of accelerated growth.** Firms that have integrated with e-commerce platforms have outperformed the non-integrated ones, benefiting from faster expansion and many more opportunities opening up.

- **Beyond business growth, digitalisation offers significant second-order benefits.** This year's survey highlights that firms integrated with e-commerce platforms experience better access to finance, international markets, and critical business information—irrespective of their size—which is significantly contributing to enterprise development.

- **Access to finance remains a persistent challenge, but it is not insurmountable.** The study finds that as firms grow, financial constraints ease. E-commerce integration helps mitigate financial barriers by fostering business expansion and creating a stronger information ecosystem for MSMEs that helps them overcome a lack of collateral.

- **E-commerce serves as a gateway to global trade, particularly for micro-enterprises.** The majority of export-oriented firms that are integrated on e-commerce platforms are leveraging e-commerce platforms to reach international markets. These include a significant percentage of micro firms also, proving that even smaller and less productive firms can access global opportunities through digital channels.

- **Looking ahead, fostering the digital transformation of MSMEs can help tap the full potential.** Enhancing technology access, upgrading skills, and developing a robust support ecosystem will enable firms to maximise the benefits of e-commerce integration. Strengthening both physical and digital infrastructure is key to unlocking the full potential of MSMEs.

EXECUTIVE SUMMARY

MSMEs dominate the enterprise landscape in India. However, they struggle to scale up because of their low productivity levels. This is due to the sector's fragmented and largely informal nature. Enterprise development is, therefore, key for MSMEs to scale up, which is a necessary for a high and sustained growth rate and employment generation in India.

Limited access to finance, markets, information, and skilled human resources are some of the key challenges impeding enterprise development in India. Digitalisation offers a pathway to address these issues and accelerate enterprise development. Although Internet usage is growing among MSMEs, they are unable to exploit its potential. This is because MSMEs, particularly micro-enterprises, cannot independently innovate or leverage the full potential of technology-based tools and platforms, and often need support from large enterprises and the government.

While some data on MSMEs' Internet usage exists, qualitative insights on digitalisation's impact on their growth and development are currently lacking. To examine the impact of digitalisation, specifically e-commerce platforms, on the growth and development of MSMEs, a survey of 2,365 Udyam-registered manufacturing sector firms was conducted in 2024. This is the third edition of the survey. A purposive sampling technique was used to compare integrated and non-integrated businesses. The survey's design facilitated both time series and cross-sectional data analysis, making it a key strength of the study. While previous surveys have established the key benefits, such as increased sales, higher profit margins, and positive effects on employment, this study also focuses on examining the second-order effects of digitalisation on MSMEs. Specifically, it explores how integration enhances access to finance, international markets (exports), and information—factors that play a crucial role in enterprise growth and development. This report presents the insights derived from the survey data.

Contrary to the perception that the MSMEs sector is struggling and underperforming, the surveyed enterprises exhibit a robust growth trajectory. Most respondents reported a rise in sales, profit margins, and hiring compared to the previous two years. About 50% of the surveyed enterprises saw an increase in turnover and profits in 2023-24, compared to 2021-22. Importantly, over time, a greater percentage of firms are moving towards higher turnover ranges, employment levels and profit margin. Even though these firms are specifically Udyam-registered, it is a promising trend, as more businesses are actively registering on the Udyam and the Udyam Assist Portal of the Government of India, indicating a positive shift.

Integrated firms outperform non-integrated ones. Our sample shows that to begin with, more productive firms join e-commerce platforms. These are largely firms that have higher turnover and profit margins. Over years, these firms have shown a stronger growth trajectory, surpassing non-integrating firms, in terms of turnover, profit and employment. Notably, a substantial share of firms (90%) reported growth in sales and profit margins after e-commerce integration. Nearly

50% of the integrated firms hired new employees post integration, out of which nearly 75% firms also hired women employees.

Scaling up business is the key reason for joining e-commerce platforms, yet offline channels remain the dominant mode of sale. A majority of integrated firms joined e-commerce platforms to boost their sales, with most enterprises starting to sell on e-commerce platforms after the onset of the COVID-19 pandemic. About 71% of enterprises joined e-commerce platforms during or after 2020. However, the survey found that while the reliance on online operations, including e-commerce platforms, is growing, offline channels remain significant medium for sale.

While previous ICRIER surveys established that e-commerce integration positively impacts business growth potential, the current study also highlighted evidence for interesting second-order effects, which can contribute significantly to enterprise development. The survey found that as a firm's size increases, its ability to access finance and the international market improves. Additionally, firms can gain improved access to market information.

The survey found that e-commerce integration positively impacts MSMEs' ability to raise external finance at relatively more favourable terms. By enabling firms to expand, e-commerce enhances the long-term prospects of MSMEs. The survey revealed that relatively larger firms in the MSME segment and those integrated with e-commerce performed better in terms of financial access. The survey found that the average interest rate on MSMEs' loan is about 12.27%. Compared to this, on an average, medium sized firms and integrated firms pay a relatively lower rate of interest (11.59% and 12.15%, respectively). Additionally, a relatively higher share of e-commerce-integrated firms could access collateral-free loans. Among integrated firms, 51% found their e-commerce presence helpful in raising external finance from banks and non-banking financial corporations, with medium (60%) and small-size enterprises (55%) doing better on this front than micro firms (48%). Additionally, 30% of integrated firms found e-commerce beneficial for raising collateral-free loans, with medium firms (33%) leading, followed by small (31%) and micro (29%) firms.

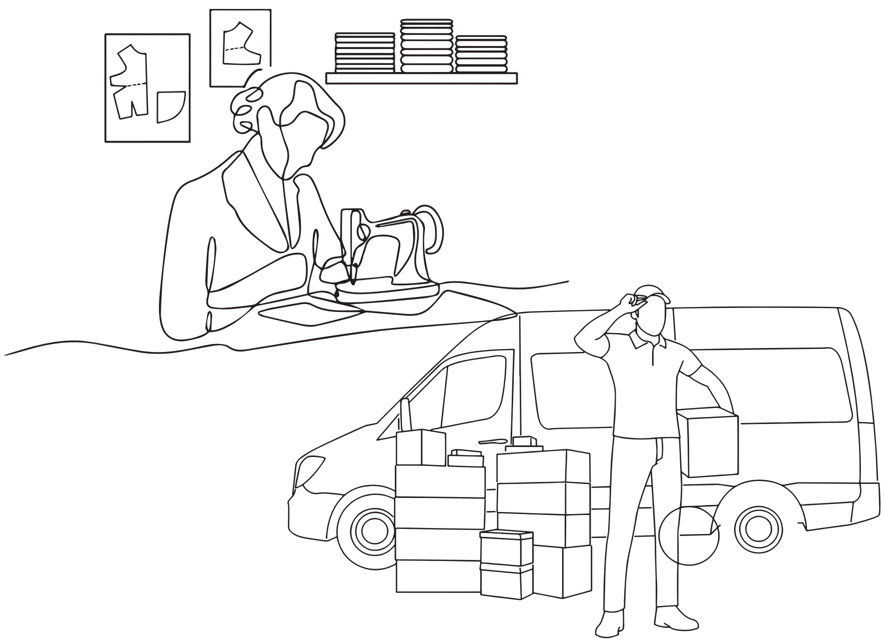
Many enterprises made new investments or introduced innovative practices after integrating on e-commerce platforms, enhancing their long-term growth and sustainability. About 62% of the integrated firms invested in training their employees, and nearly 50% invested in the R&D of new products. More than 90% of the firms also introduced innovative practices after integrating on platforms. The survey found that while business growth remains the most important benefit of integrating on e-commerce platforms, medium-size firms value qualitative benefits (such as access to market insights and improvement in product quality) more than micro- and small-sized firms.

As the firm size increases, its export potential too, rises, and integrated export-oriented firms use e-commerce platforms extensively to export their products. About 16% of all surveyed enterprises export their products, with medium-size enterprises having a higher share of export-oriented firms. The survey found that e-commerce offers firms, especially micro-enterprises, a pathway to participate in global trade. About 93% of the export-oriented, integrated firms rely on e-commerce platforms to export their products to international markets. This includes a considerable share of micro-enterprises, highlighting that even relatively small firms can utilise e-commerce to reach global markets if they choose to export.

Importantly export potential can be tapped better through e-commerce, which can address some conventional issues that firms face in exporting their products, like information access. Nearly 21% of the firms that do not currently export believe their products have export potential. More integrated firms (24%) recognise this potential than non-integrated ones (18%). Some of the most prominent challenges in accessing international markets include information gaps and limited access to finance. About 38% of the respondents reported that integrating on e-commerce platforms can help them overcome some of these challenges.

While market access and financing are major hurdles for non-integrated firms, the primary challenges faced by integrated firms are notably distinct, necessitating a two-pronged approach to policy intervention. Non-integrated firms continue to face the conventional issues, including access to markets and finance. Compared to this, top five challenges ranked by integrated firms included unavailability of skilled manpower to run online business, targeting right customers online and marketing products, among others. Thus, while it appears that e-commerce has been able to subdue the conventional issues to an extent, there is a need to address others, which are affecting firms ability to fully leverage platforms. Notably, nearly 48% of integrated MSMEs suggested that there is a need to address errors and delays like account de-listing, and logistics/payment delays, about 44% of firms suggested that e-commerce companies should work towards checking/reducing frauds.

Going forward, it is essential for the government create a supportive digital ecosystem that empowers MSMEs to leverage e-commerce opportunities. Taking measures to strengthen cybersecurity, enhancing access to technology and bridging information gaps will be key to unlocking the sector's untapped potential. Moreover, fostering an ecosystem enabling firms to maximise the second-order benefits of e-commerce integration is important. This can be achieved by developing supportive physical and digital infrastructure that harness the advantages of e-commerce integration. By addressing these challenges, India can further accelerate enterprise development, ensuring that MSMEs continue to play a vital role in economic growth and employment generation.



Annual Survey of Micro, Small and Medium Enterprises (MSMEs) in India

The Role of Digitalisation in Enterprise Development

1. Introduction

Once considered vestiges of the informal sector, micro, small and medium enterprises (MSMEs) now dominate India's enterprise landscape.[1] In 2023-24, there were about 73.4 million unincorporated non-agriculture enterprises in the country, employing 120 million people and contributing 16.5% of the total gross value added (GVA).[2] There has been a 12.8% growth in their number of and a 10% growth in their employment count in 2023-24, compared to 2022-23.

To formally recognise MSMEs in India, a registration process was introduced, now known as Udyam registration.[3] There are over 35 million firms registered on the Udyam Portal and over 25 million firms registered on the Udyam Assist Portal, which aims to formalise the informal sector enterprises. These together employ more than 250 million people.[4] MSMEs contribute significantly to the national output, employment and play a key role in India's economic development.

Yet, MSMEs in India grapple with the issue of low productivity. Existing studies highlight that productivity levels in MSMEs are about half that of large enterprises, with the problem more pronounced in emerging market economies.[5] In India, small businesses are only one-fourth as productive as large enterprises, exhibiting a wider gap than that in most emerging market economies.[6] Larger scale generally raises productivity on account of improved access to intangible capital, global connections, and better access to financial capital.[7]

The MSME sector is mainly informal and fragmented, and so productivity levels tend to be low. Micro-enterprises comprise the largest share of India's MSME sector. About 85% of unincorporated firms are own-account enterprises and more than 98% of the Udyam registered enterprises are micro-sized. The share of registered enterprises was only around 37% of the total unincorporated non-agriculture enterprises in 2023-24. In the manufacturing sector, however, the share of registered enterprises is much smaller.[10] These structural characteristics inhibit the enterprises' potential to scale up, without which they cannot contribute to the national output and employment in a sustainable manner.

Enterprise development, therefore, is key to India's growth and employment generation. While there is no universally accepted definition of 'enterprise development', it can be understood to mean "the act of investing time and capital to help people establish, expand or improve business".[11] It involves establishing a new enterprise or scaling up an existing business to become a key source of economic growth, employment, and wealth creation.[12] In India, where there is a predominance of small and informal-sector firms, enterprise development initiatives can help firms scale up and improve their productivity, thereby boosting their output and employment generating potential.[13] To achieve this, tackling the fundamental challenges hindering enterprise development is essential.

Limited access to finance, markets, information, and skilled labour are some challenges impeding enterprise development in India.[14] Access to finance is a significant bottleneck. Financial institutions perceive MSMEs as high-risk borrowers due to a lack of data on creditworthiness.[15,16] Globally, 79% of loans require collateral. MSMEs typically do not have fixed assets, machinery, or equipment.[17] On average, fixed assets owned per establishment were valued at INR 318,000 in 2022-23 for unincorporated non-agriculture enterprises, and the value was lower (INR

215,000) when only manufacturing enterprises were considered.[18] In India, unsecured loans accounted for only a quarter of the total credit disbursed by scheduled commercial banks (as of March 2024); private banks mainly lent this at relatively high-interest rates.[19] As a result, MSMEs face higher borrowing costs than larger firms due to the higher risk premiums charged by lenders, exacerbated by informational asymmetries that increase banks' risk perception.[20,21]

MSMEs also lack adequate access to information and markets, particularly international markets. MSMEs struggle to capitalise on market opportunities due to their limited access to market information, the high costs of accessing such information, and other resource constraints. Additionally, trade barriers, including standards, technical regulations, logistics, and administrative procedures, disproportionately affect MSMEs.[22] Their participation in global trade is further hindered by asymmetric knowledge and a lack of understanding of foreign markets.[23,24]

Digitalisation can help address some of these issues and accelerate enterprise development. Digitalisation is a key driver of enterprise development and an important tool to address enterprises' structural challenges. Digitalisation brings operational efficiencies for SMEs, directly supporting value creation and allowing these firms to tap opportunities, enhance performance, and embrace innovation and competitiveness.[25] Connectivity, Internet usage, and digital integration help SMEs become sustainably competitive vis-à-vis larger firms.[26,27] Additionally, digital technologies foster entrepreneurship by optimising processes, improving decision-making, and enabling product customisation.[28] Therefore, enterprises must embrace digitalisation and establish an online presence, including developing websites, to enhance visibility and recognition. Digitalisation promotes internal, systemic efficiencies, and can also improve external linkages, such as improved access to markets and finance, which can contribute to enterprise development.[29]

Although Internet usage is growing among unincorporated-sector enterprises, its functions remain limited. The use of the Internet for entrepreneurial activities, such as online transactions, order placement, and UPI payments, was about 26.7% in 2023-24 for rural and urban areas combined.[30] In 2022-23, most firms used the Internet for banking purposes (615 per 1000 establishments), followed by to access other financial services (236 per 1000). A minute proportion of firms used the Internet to make sales (81 per 1000) and deliver products (27 per 1000). This suggests that MSMEs are currently not optimally using the revenue generating potential of digital interventions that can significantly contribute to enterprise development.

Low usage is because MSMEs, particularly micro-enterprises, cannot independently innovate or leverage full potential of technology-based tools and platforms and often need external support to leverage technology. Smaller businesses are often slow to embrace digital technologies due to low awareness, limited internal resources, skill gaps, and financial constraints. Digitalising the vast and heterogeneous MSME sector requires collaboration between the various stakeholders. This can also create an enabling environment for enterprise-level innovation, resource sharing, skill development, and access to financial and policy support. MSMEs require such assistance, including access to common facility centres for manufacturing, mentoring, and financial assistance, particularly from large enterprises and government, for their skill development and digital transformation. This is where e-commerce platforms can play a role.

While some data on MSMEs' internet usage exists, qualitative insights on digitalisation's impact on their growth and development are lacking in the current statistical system. Although the benefits of digitalisation for small businesses are commonly acknowledged, there is limited data to support these claims. To bridge this gap, ICRIER surveyed 2,365 MSMEs across India to assess the impact of digitalisation on their business prospects, access to finance, and export potential. The survey explored how digital interventions, particularly participation in e-commerce platforms, influence enterprise development. This report presents the insights derived from the survey data.

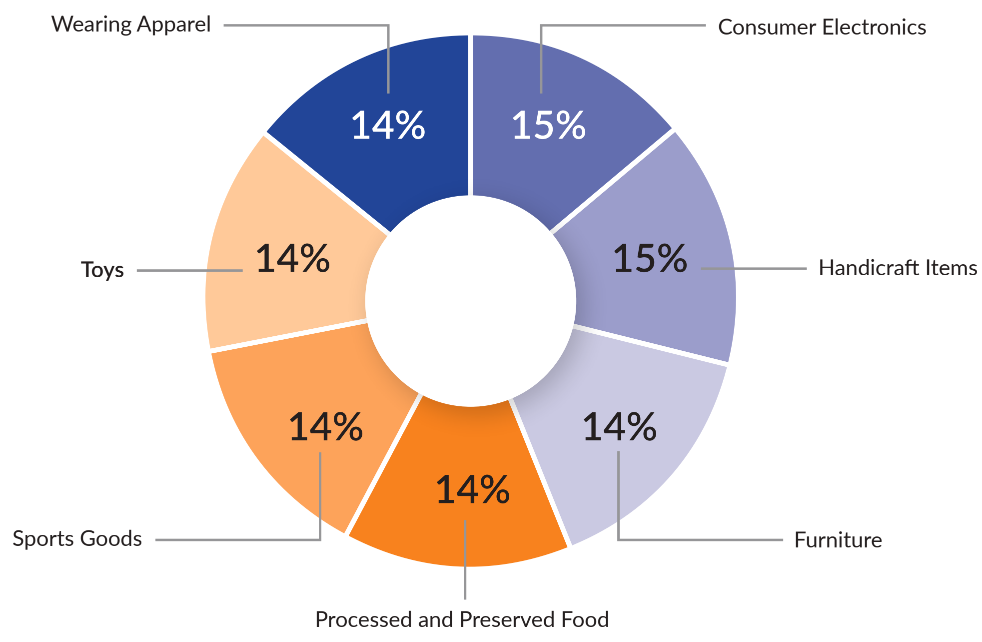
The rest of the report is organised as follows. Section 2 introduces the survey by describing the sample framework and data collection process. Section 3 presents the characteristics of the sample, such as age, ownership and turnover of the enterprise. Section 4 discusses the typical journey of the enterprise, focusing on integrated firms. It presents how business and growth prospects emerged after firms integrated on e-commerce platforms. Section 5 discusses MSMEs' experience accessing formal finance and compares the experiences of integrated and non-integrated firms. Section 6 presents the findings on international market access, particularly the export orientation of MSMEs and contrasts between integrated and non-integrated enterprises. Section 7 highlights the challenges and recommends a way forward.

2. Survey Framework and Data Collection Process

The objective of the survey was to examine the impact of digitalisation, specifically e-commerce platforms, on the growth and development of MSMEs. A total of 2,365 Udyam-registered manufacturing-sector MSMEs were surveyed for this study. Purposive sampling was followed to include an equal number of firms integrated on e-commerce platforms and those that are not. A pilot survey was conducted in the National Capital Region in July 2024, and the field survey was conducted between September and November 2024.

A cluster-based approach was adopted to identify MSMEs within the selected sectors. The study covered seven sectors: ready-to-wear garments; furniture; processed and preserved food; toys; sports goods; handicraft products; and consumer electronics (Figure 1). An equal number of respondents were selected from each product category to ensure that sectoral biases do not impede the findings.

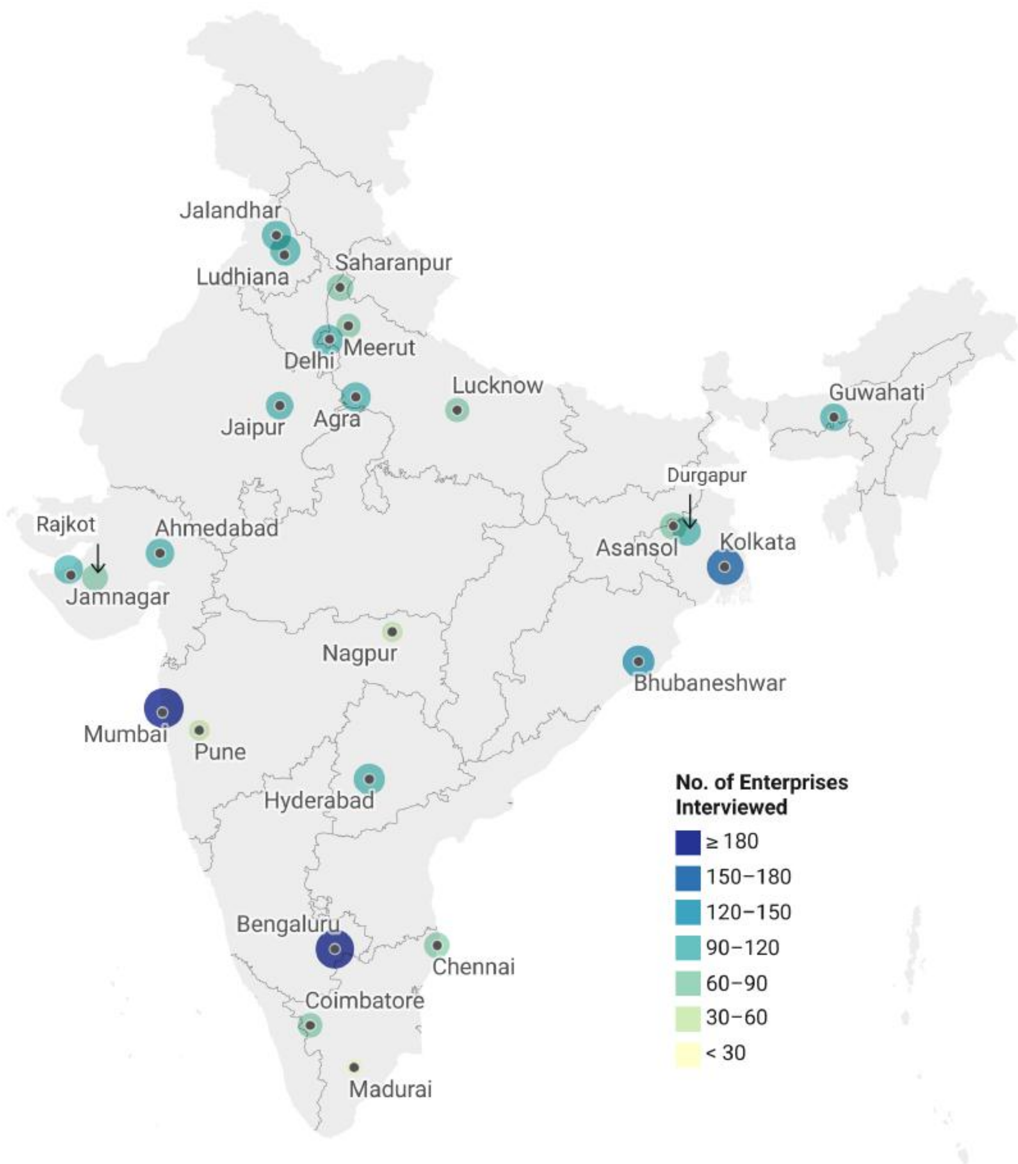
Figure 1: Distribution of enterprises by sectors (in %)



Source: *Authors' computation of data from ICRIER's MSME survey*

The selection of states and cities was as per the previous survey, which was based on a natural fallout of majority clusters in selected product categories (Figure 2). Twelve Indian states and 24 cities/centres were covered in the survey.

Figure 2: Geographical distribution of MSMEs surveyed



Source: Authors' computation of data from ICRIER's MSME survey

Although a purposive sampling methodology was used, the sample reflects key characteristics of India's MSME landscape. Notably, while only Udyam-registered enterprises were selected for the survey, the Annual Survey of Unincorporated Sector Enterprises (ASUSE) shares similarities with the sample, particularly in MSME concentration in Uttar Pradesh, West Bengal, and Maharashtra. Furthermore, key manufacturing-sector products with a significant MSME presence sold online, including apparel, food products, furniture and handicraft, were selected for this study.[31]

Handicraft manufacturing unit in Srinagar, Jammu and Kashmir

Instead of concentrating on unincorporated-sector enterprises, which have unique characteristics, certain inherent biases, and limited comparability, this study focused on Udyam-registered firms as they offer a more consistent basis for comparing firms. Approximately 98% of Udyam-registered firms are micro-sized, 2% are small, and around 1% are medium. While the overall pattern is similar in the study's sample, small and medium-size firms are adequately represented.

Based on prior experience, secondary research, and exhaustive local information, multiple industrial belts/clusters were identified in each city using purposive sampling. Once the cluster was identified, a known landmark was selected as the 'starting point' for applying the right-hand rule method to identify MSMEs within a selected cluster. The investigator then began visiting the MSMEs located on the right side of the starting point to ensure no MSMEs are left from within the identified sectors in the selected clusters.

Screener questions were used to determine the eligibility of the targeted respondents. Only the eligible respondents were surveyed. Structured questionnaires were used to conduct interviews. The questionnaire was translated into the language most commonly spoken in the respondent's state. Language experts carefully assessed the translations to ensure the questions retained their original meanings. Show cards and a 'definition sheet' were shown to respondents, wherever required, along with the questionnaire.

To make appointments with respondents, interviewers sent emails, made telephonic calls, or physically visited their premises. Based on the respondents' approval, interviewers either sought an immediate interview or an appointment to capture their responses later. Only the owner or key decision-maker of the MSME was interviewed. All surveys were conducted face-to-face and lasted about 30 minutes each. A computer-assisted personal interviewing device was used to record the responses. All responses were geo-tagged.

The survey questionnaire was divided into five sections:

The first section was designed to capture the enterprise's innate characteristics, including its ownership structure, the education level, age, and gender of the owner, and age of the enterprise. A firm's inherent characteristics are crucial to adopting digital technologies.[32] Moreover, this information is useful in creating a profile for the enterprise and cross-tabulating data on the firms' characteristics to examine trends by firm size, ownership, age, and other factors.

The second section documented firm performance, emphasising business growth, investments and innovation. It included questions on the performance of the enterprises over time in general, and post-integration. All these aspects contribute to enterprise development and its potential to scale up. Data from the past three years is presented in the report to track the firm's progress across key indicators over time. Additionally, this section delved into the motivations for adopting e-commerce, as well as the reasons for not doing so.

Section three of the questionnaire focused on the firms' access to finance. It is well-acknowledged that MSMEs face challenges in accessing formal finance, adversely impacting their potential to scale up and grow. Firms were asked about their sources of finance, interest rate paid, the issue of collateral, and, more generally, challenges faced in accessing formal finance.

Section four examined the firm's export orientation and potential to access international markets. The previous survey found that the more productive firms integrate with global trade as they are more competitive. This year's survey dwelled deeper into firm's potential to export, the challenges faced, and whether e-commerce platforms enable greater trade integration.

The fifth section of the questionnaire focused on key policy considerations. It examined the challenges encountered by both integrated and non-integrated MSMEs, explored the support and assistance they receive or utilise, and gathered their suggestions for improvement.

Although the primary survey was conducted in person, and the large sample size means the findings are largely conclusive, certain limitations remain. First, the responses are based on the self-declaration of respondents and without verification through books of accounts or other published data. Second, as a purposive sampling technique was used, the findings cannot be generalised beyond a specific homogeneous group. Third, as previously mentioned, the survey focused solely on Udyam-registered firms, meaning it does not account for the informal sector, where a significant portion of MSMEs operate. However, the survey offers valuable insights, providing a direction for the outcome and contributing to a more comprehensive understanding of the subject.

Indian MSMEs: The story of recovery, resilience, and resurgence.

Tracing the journey of the same firm over the last three years a promising picture emerges, which does not conform to the impression in the public narrative that MSMEs are largely in distress. We find that a substantial share of firms experienced an increase in turnover, profit margins and hiring. The integrated firms among these out-performed the non-integrated ones. It is, however, worth mentioning that only Udyam registered firms were surveyed for this study.

Most of the firms reported a growth in their turnover over the last three years, while a miniscule share of firms saw a decline. About 50% of the surveyed enterprises saw an increase in turnover in 2023-24, compared to 2021-22. In comparison, only 2% of the surveyed enterprises saw a decline, while others stayed within the same range. While an equal share of integrated and non-integrated firms saw an increase in the turnover, the absolute level of increase in the size of the turnover was higher for the integrated firms. This is primarily due to the expanded market reach gained through platform integration. Interestingly, the survey revealed that most enterprises joined e-commerce platforms to expand their market reach, and the majority successfully achieved this goal.

Additionally, nearly 46% of firms experienced an increase in their profit margins in the last two years. More than 50% of the firms that experienced a rise in profits are integrated on e-commerce platforms. Moreover, the absolute margin of increase is higher for integrated enterprises than for non-integrated firms.

Notably, around 20% of the surveyed enterprises moved from lower to higher employment ranges - that is - from hiring less than 10 employees to hiring between 10-50 employees, reflecting a clear increase in their overall hiring. A relatively higher percentage of integrated firms reported such a shift, compared to non-integrated ones. During the survey, the authors found that as firms expand their business, they tend to hire more production workers, but on a contractual basis. This is particularly true for firms that are export oriented. When they receive large export orders, they increase the number of production workers on a piece-rate basis. The pilot interactions revealed that in some sectors, like ready-made garments and food processing, these are largely women. Anecdotally, it was reported that on production floors, women are more efficient than men as they take fewer breaks, are more punctual, and work more diligently.

Pictures of the production floors of two units from pilot interviews conducted in September 2024

3. Characteristics of the Sample

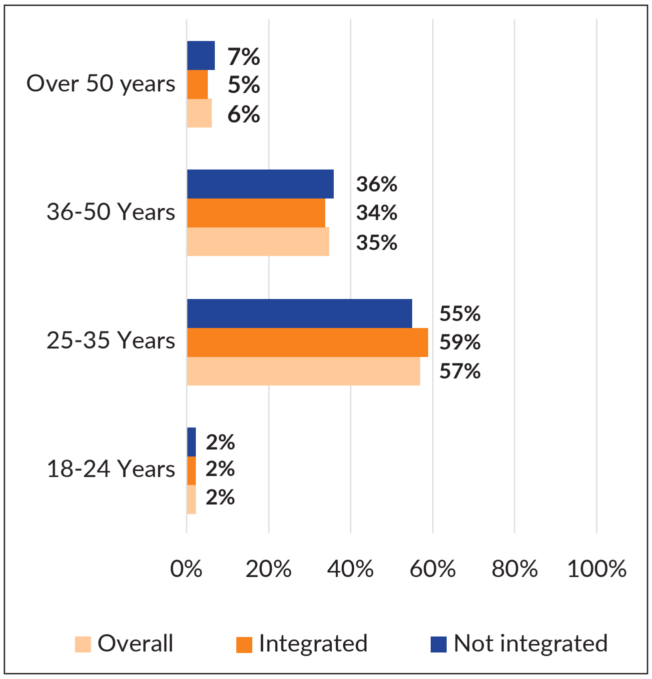
Table A1 in the Appendix presents a snapshot of the characteristics of the firm and its owner. It includes information about the owner's age, gender, and education. It also provides information on the enterprise's age and whether it operates out of a business or household cluster.

Most smaller-sized enterprises are relatively new, while larger firms have been operational longer. Over 50% of the enterprises were established after 2010. Around 12% of medium-sized enterprises have been operating since before 1990, compared to 9% of small-sized firms, and 7% of micro-enterprises.

Most firms operate within business clusters, with relatively larger-sized firms having a higher presence in these clusters than household clusters. Enumerators noted whether firms operated within a business or household cluster. Enterprise growth prospects improve in business clusters due to better supporting infrastructure. About 78% of all enterprises and 100% of medium-sized firms operated within a business cluster. Consequently, a relatively larger percentage of micro firms (about 28%) were operating within household clusters.

Integrated firms are typically owned by younger, more educated individuals than non-integrated ones. Figure 3 shows that 61% of integrated firm owners and 57% of non-integrated firm owners are under 35 years old. About 85% of integrated enterprise owners and 81% of non-integrated firm owners hold graduate-level or higher educational qualifications.

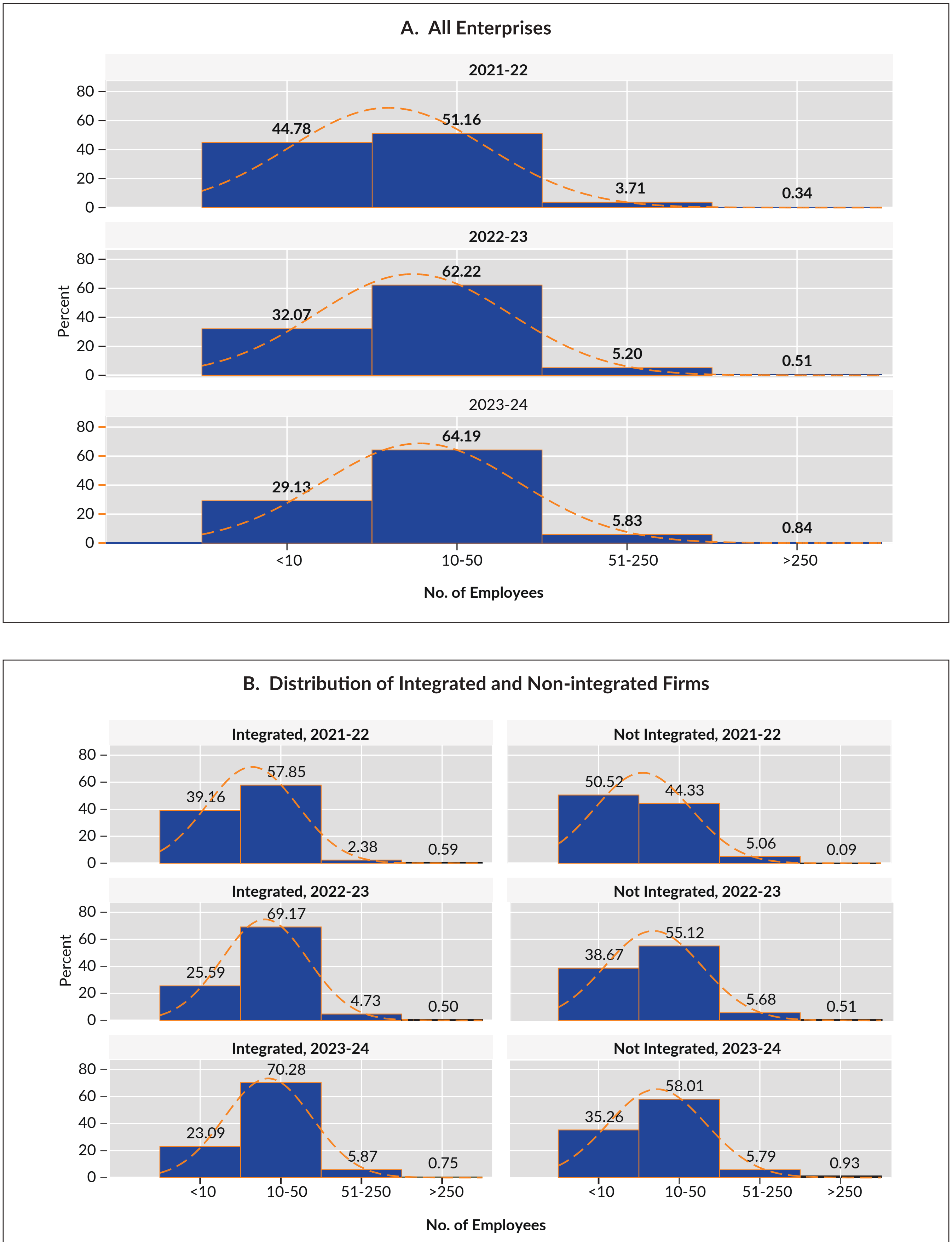
Figure 3: Distribution of enterprises by age of owner and e-commerce integration (in %)



Source: *Authors' computation of data from ICRIER's MSME survey*

A greater percentage of integrated firms have employees in higher employment brackets, and the share of these firms has risen over time. In general, there has been an increase in the share of firms with a higher number of employees over the years, thereby indicating a growth in hiring in MSMEs. Figure 4 illustrates that integrated enterprises increasingly hire more employees than non-integrated ones. In 2021-22, 61% of integrated enterprises employed more than 10 workers, compared to 49% of non-integrated enterprises. By 2023-24, this proportion rose to 77% for integrated and 65% for non-integrated enterprises. This indicates that integrated enterprises not only employ a larger workforce but also exhibit a faster pace of hiring compared to their non-integrated counterparts.

Figure 4: **Distribution of enterprises by number of employees from 2021-22 to 2023-24 (in %)**



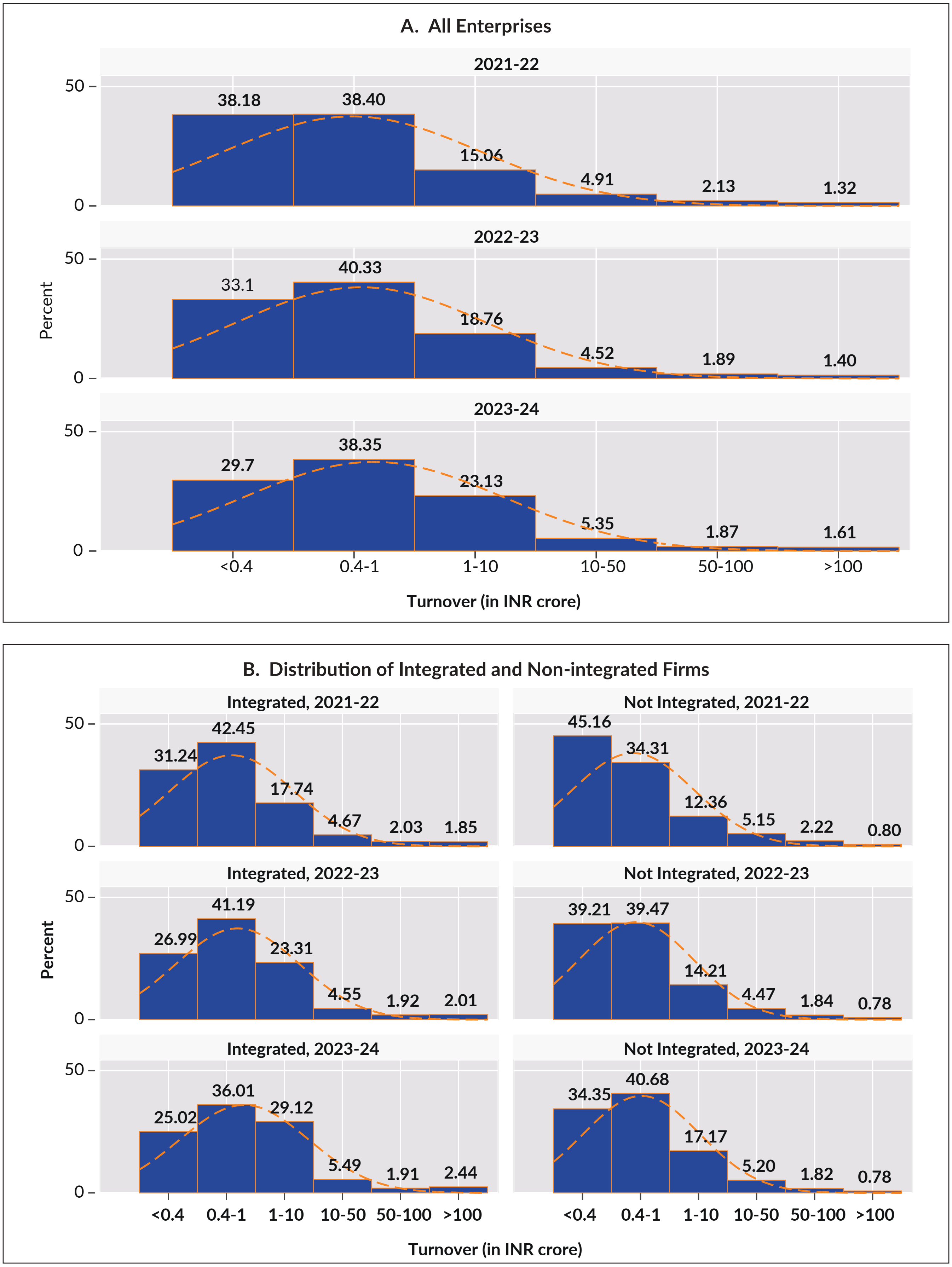
Source: Authors' computation of data from ICRIER's MSME survey

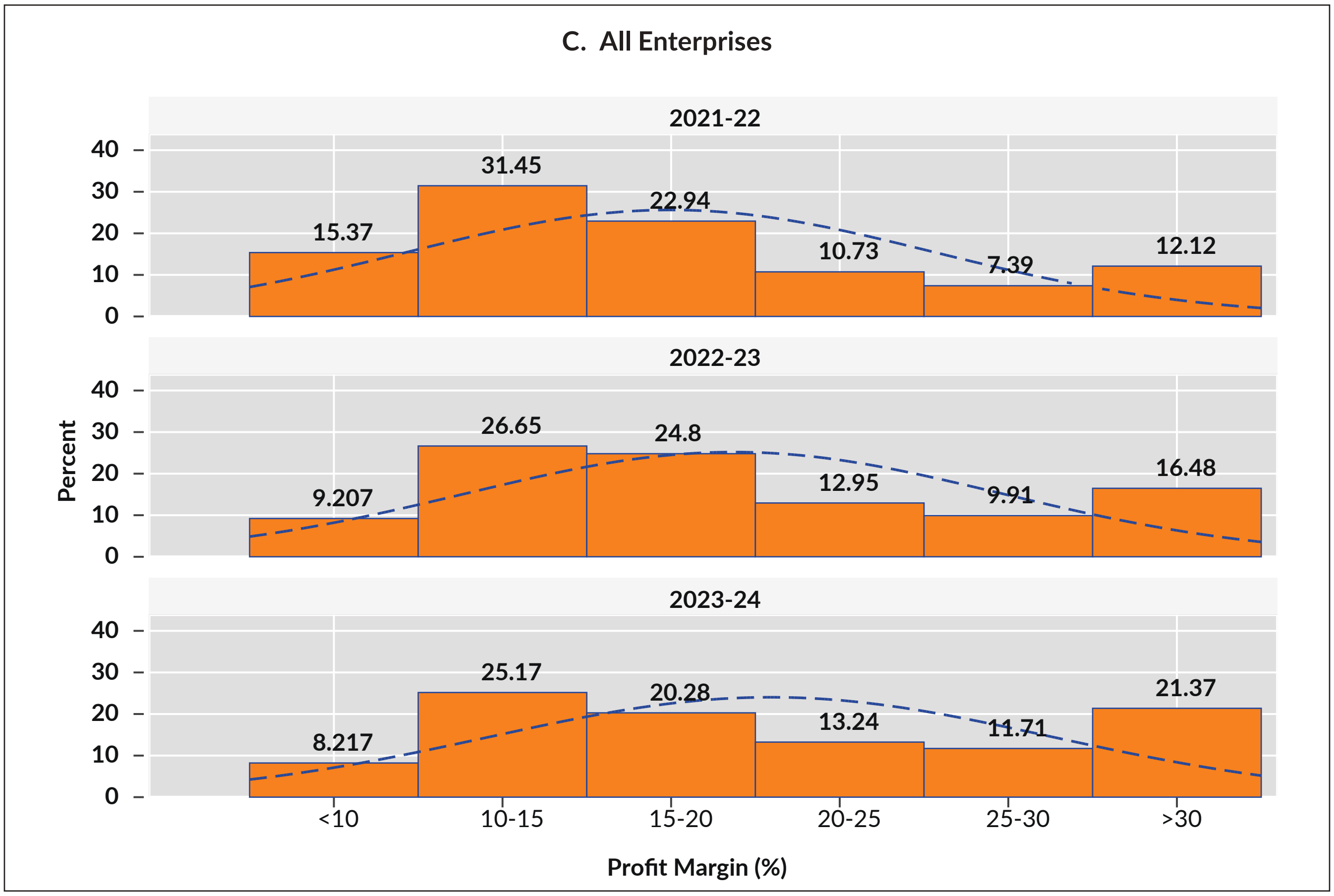
While most firms have more than half of their employees as permanent staff, a relatively higher share of integrated firms has more permanent employees than non-integrated ones. This is driven by the needs of online operations. About 47% of integrated firms and 42% of non-integrated firms have more than 50% permanent employees. Interestingly, smaller-sized firms (micro and small enterprises) have a relatively higher share of permanent employees, possibly because their total employee base is small. Nearly 70% of the medium-sized enterprises have less than 50% permanent employees, compared to 61% of small enterprises and 52% of micro-enterprises.

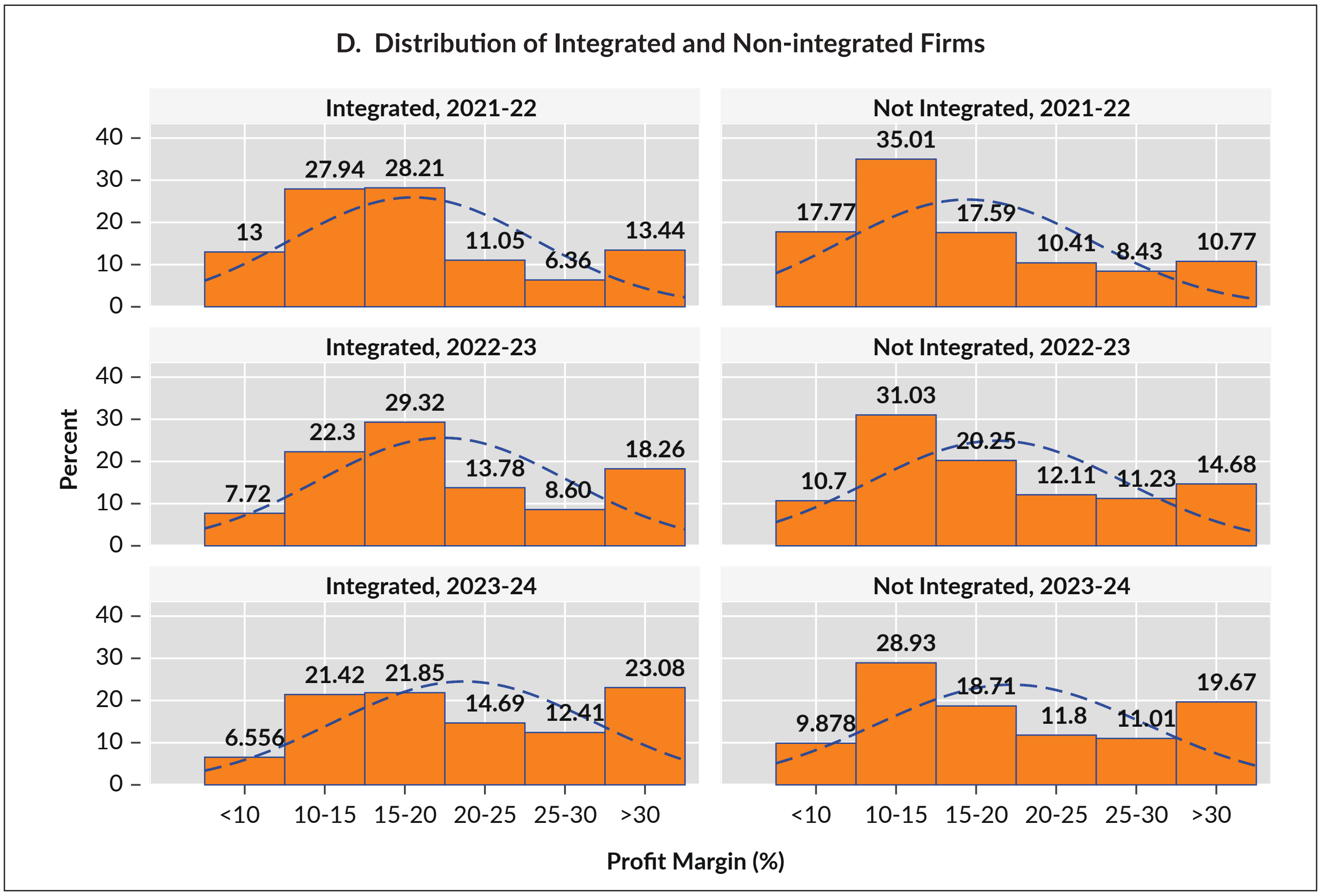
Surveyed MSMEs have demonstrated a growth trajectory, with a consistent increase in turnover over the last 3 years. Generally, the proportion of firms in higher turnover ranges has risen over the past three years, demonstrating the growth of MSMEs. The estimated average turnover for the surveyed MSMEs increased by 11.4% from 2022-23 to 2023-24.[33] When this is compared to the unincorporated sector firms, it is observed that the surveyed enterprises, all of which are Udyam registered, are doing significantly well. The unincorporated manufacturing establishments reported a fall of 6.7% in their gross value added in 2023-24 compared to the previous year.[34]

Integrated firms have outperformed non-integrated enterprises. A greater percentage of integrated firms demonstrated growth. As seen in Figure 5 (A and B), in 2021-22, about 25% of integrated enterprises reported a turnover exceeding INR 1 crore, compared to 20% of non-integrated enterprises. By 2023-24, this share increased to 38% for integrated enterprises and 25% for non-integrated ones. Thus, over time, there has been a shift toward higher turnover ranges, with a greater proportion of integrated firms reflecting this trend. The estimated average turnover for integrated firms increased by 13% between 2022-23 and 2023-24 as compared to 8.5% increase for non-integrated firms. Importantly, the share of enterprises with over INR 5 crore turnover increased from 11% in 2021-22 to 13% in 2023-24 for integrated category, while it remained constant at 10% for non-integrated firms.

Figure 5: Distribution of enterprises by turnover (INR crore) and profit margins from 2021-22 to 2023-24 (in %)







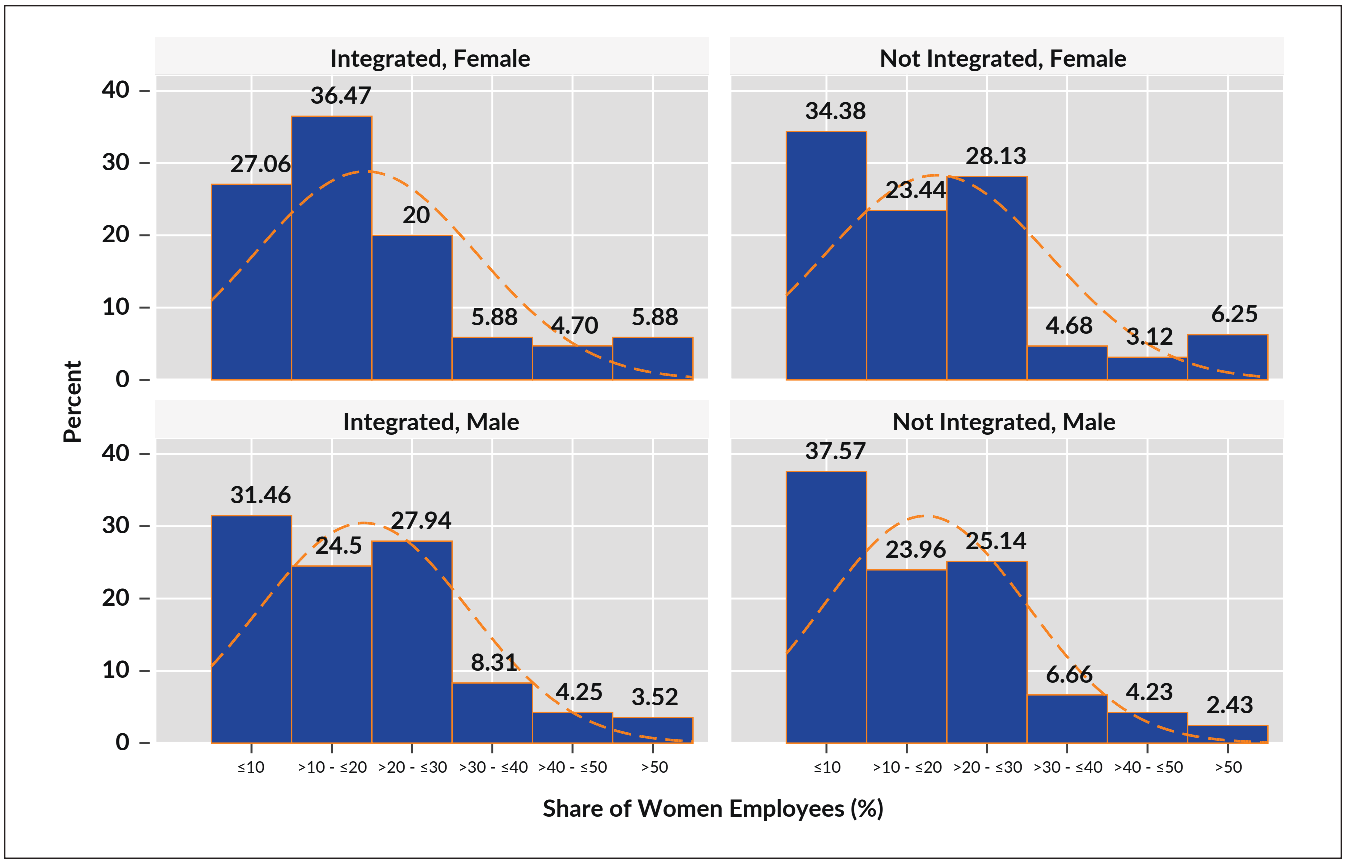
Source: *Authors' computation of data from ICRIER's MSME survey*

Profit margins have increased over the years, again, with integrated firms outperforming non-integrated ones. Most enterprises reported profit margins between 10% and 20%, while a higher proportion of small firms had margins exceeding 20%, followed by micro and medium enterprises. Over time, more firms have moved toward higher profit margins, a trend more pronounced among integrated firms. In 2023-24, a greater share of integrated firms reported profit margins above 20% compared to non-integrated firms (Figure 5, C and D).

The share of women-owned enterprises is generally low, however, among women-owned enterprises, e-commerce seems to be a preferred channel for sale. About 6% of all enterprises were owned by women. The share of women-owned enterprises was slightly higher for e-commerce integrated firms (7%) than non-integrated ones (5%). On examining just the women-owned enterprises it is seen that about 57% are integrated on e-commerce platforms compared to 50% of male-owned firms. Importantly, contrary to the view that ownership is only indicative, women continue to be the key decision-makers in the firms owned by them, including in integrated firms.

A majority of the firms continue to hire more men than women. Most enterprises do not have an adequate representation of women in their workforce, irrespective of the size or e-commerce integration. Nearly 60% of the firms have less than 20% women in their workforce. Only 2-5% of the firms have more than 50% women employees. The share of women employees is slightly higher in women-owned enterprises and among integrated firms. Yet, none of the firms surveyed had an all-women workforce, irrespective of size, sector, ownership (male or female), and status of integration (Figure 6).

Figure 6: Distribution of enterprises by percentage of women employees in the total workforce, classified further by primary owner's gender and e-commerce integration (in %)



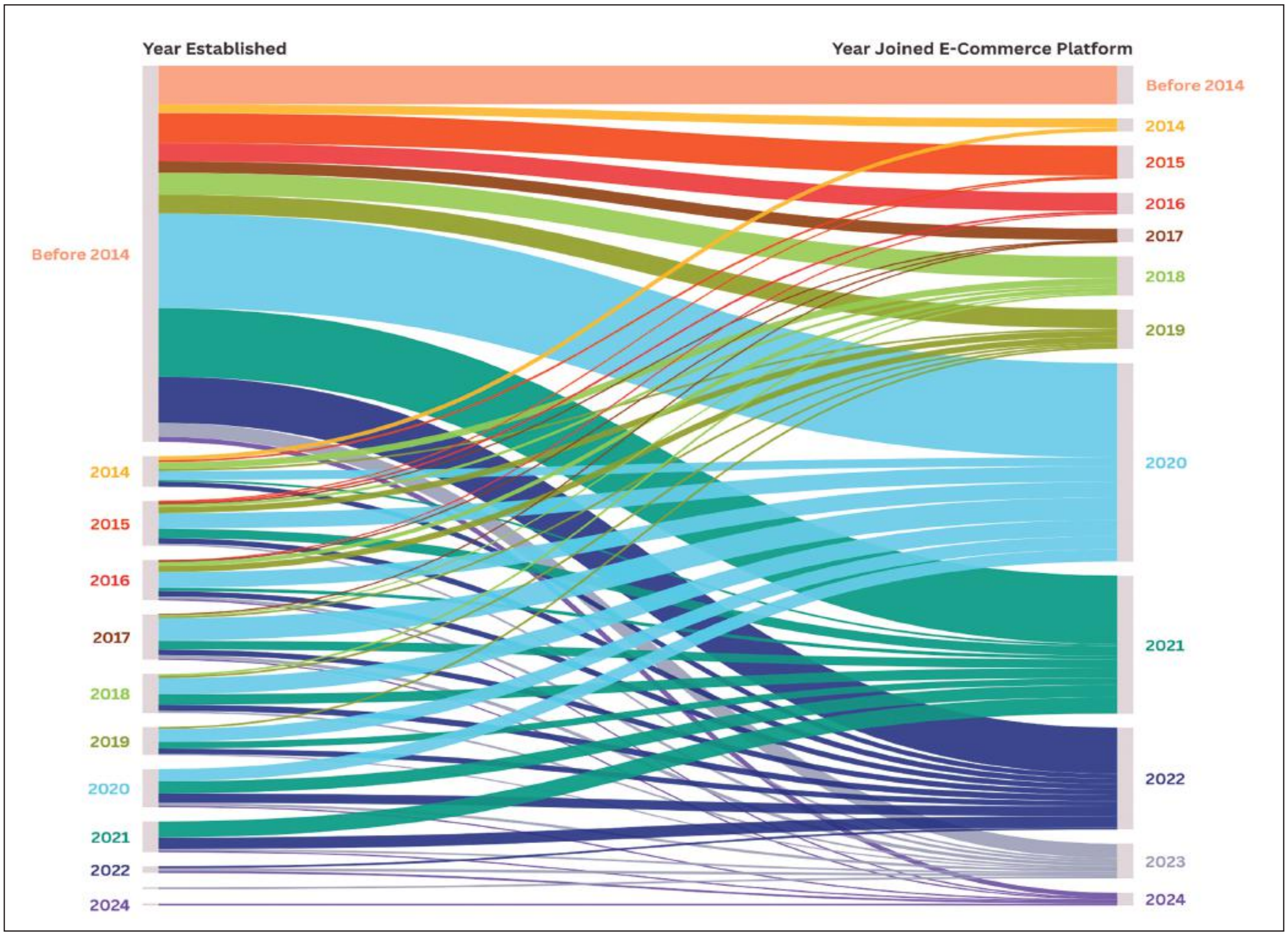
Source: Authors' computation of data from ICRIER's MSME survey

4. Leveraging E-commerce Platforms

A majority of integrated firms joined e-commerce platforms to boost their sales. About one-third of the surveyed enterprises reported they joined the platforms to improve their sales. It is widely acknowledged that e-commerce platforms allow MSMEs to expand their sales by providing an additional channel for connecting with potential customers. A relatively large percentage of medium-size enterprises reported this as the reason for joining e-commerce platforms. Firms, especially among micro-size category, said that it helped that most platforms do not have a joining fee.

Most enterprises started selling on e-commerce platforms after COVID-19. Figure 7 highlights that about 71% of enterprises joined e-commerce platforms during or after 2020, and more than 90% joined over the last decade. Notably, only 7% of the firms started selling online the same year they established operations. About 21% started selling within two years of establishing their operations. Nearly 60% of the surveyed enterprises started selling on e-commerce platforms five years after establishing their operations. About 57% of the enterprises directed 50% or more sales to business-to-business operations (or public procurement) and, therefore, selling directly to consumers was not a primary business model for them.

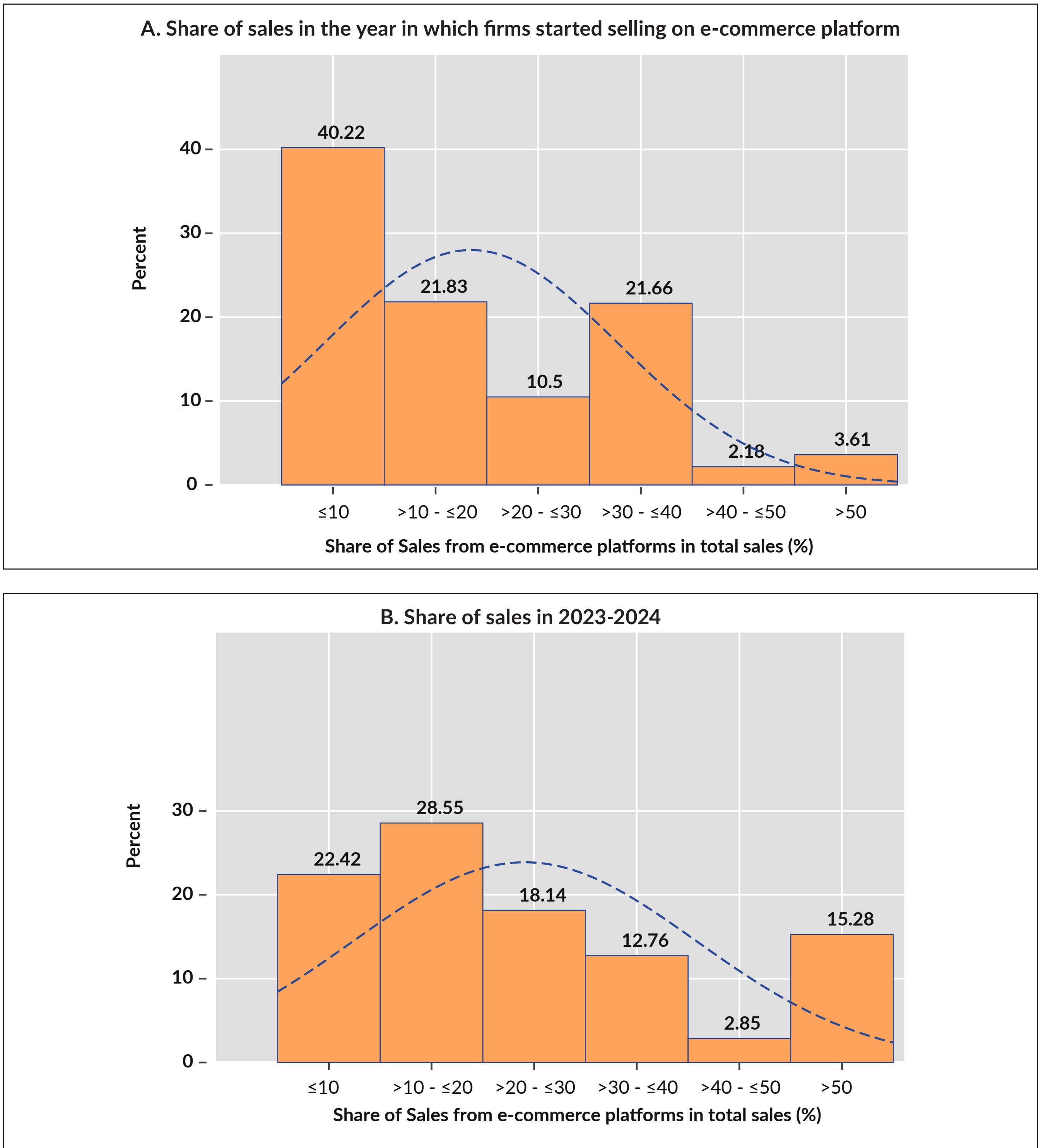
Figure 7: **Distribution of enterprise by their year of establishment and year the firm started selling through e-commerce platform (in %)**



Source: Authors' computation of data from ICRIER's MSME survey

While the reliance on online operations, including e-commerce platforms, is growing, offline channels remain dominant. Only 3% of firms initially conducted over 50% of their sales online, but this figure rose to 32% by 2023-24. Similarly, as illustrated in Figure 8, just 3.6% of firms sold over half of their output via e-commerce platforms when they first adopted them, increasing to around 15% in 2023-24. Despite the expanding role of online sales in market access, offline channels continue to be the primary mode of enterprise sales. The survey found that in 2023-24, for approximately 70% of the firms a majority of their business transactions remained offline and for nearly 85%, a majority of the transactions occurred outside e-commerce platforms.

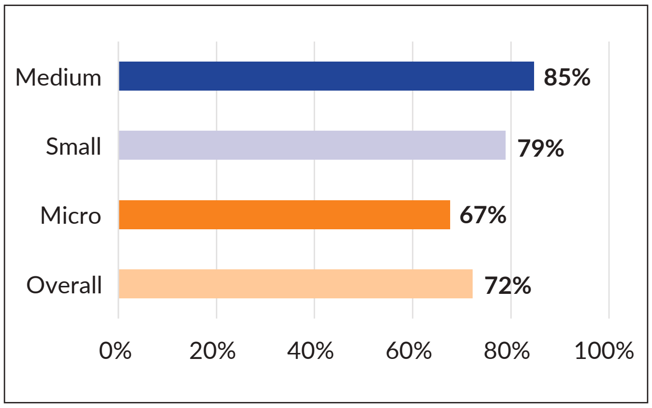
Figure 8: Distribution of enterprises by share of sales from e-commerce platforms in total sales (in %)



Source: Authors' computation of data from ICRIER's MSME survey

Most firms reported a rise in e-commerce sales compared to the previous year, and the majority plan to expand further in the next two years, indicating untapped potential. Nearly 60% of firms experienced growth in e-commerce sales, with medium-sized enterprises seeing the highest increase, followed by small and micro firms. This suggests that firm size influences business prospects after integration. When asked about future expectations, over 70% of firms indicated plans to boost e-commerce sales, with medium and small firms showing greater intent than micro-enterprises (Figure 9). Most businesses aim to increase their e-commerce sales share by up to 25% over the next two years.

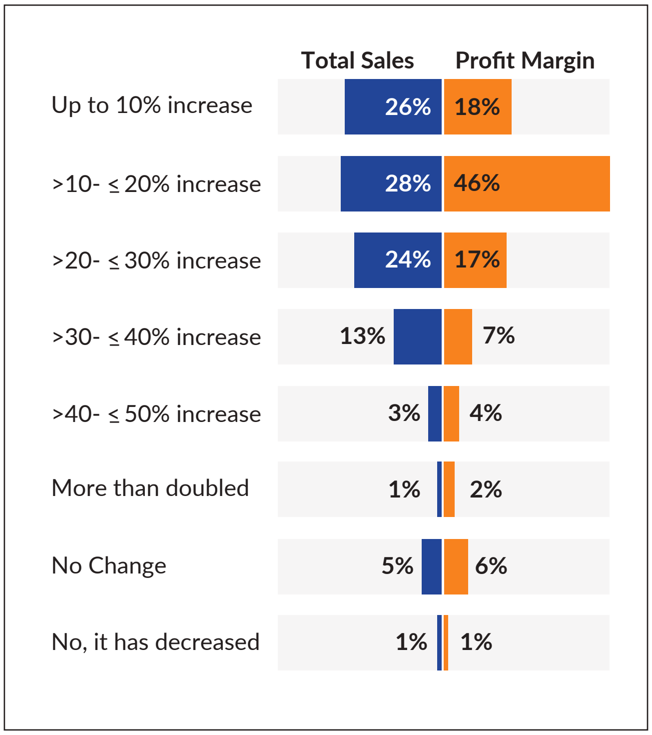
Figure 9: **Distribution of enterprises by firm planning to increase e-commerce sales (in %)**



Source: Authors' computation of data from ICRIER's MSME survey

A substantial number of firms (over 90%) reported a growth in sales and profit margins after e-commerce integration (Figure 10). The range of increase is relatively higher for small-sized enterprises for both profit margins and sales growth. Moreover, a relatively large percentage of micro-enterprises (~95%) saw increased sales and profit margins post-integration. In comparison, a larger percentage of medium-sized enterprises (about 15%) did not see any change in sales or profit margin after integration. E-commerce integration appears to have more significant benefits for relatively smaller-sized firms by allowing them greater market access post-integration.

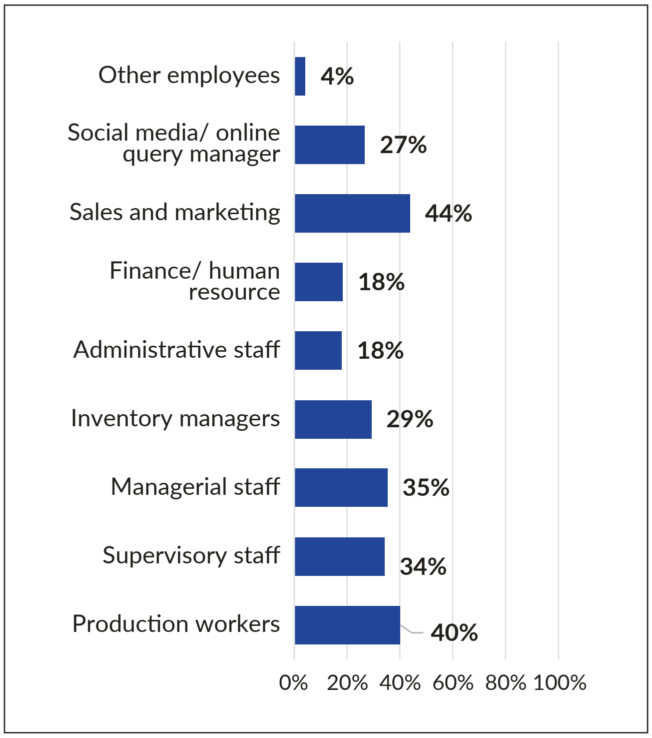
Figure 10: **Distribution of enterprises by reported growth in total sales and profit margin since integrated with e-commerce platforms (in %)**



Source: Authors' computation of data from ICRIER's MSME survey

Nearly 50% of the integrated firms hired new employees post-integration, with most also hiring women employees. These include both permanent and contractual employees. Hiring was relatively more for small-sized enterprises (57%), followed by medium (53%) and micro-enterprises (51%). About 44% of the surveyed enterprises hired new employees for sales and marketing positions to run the e-commerce operations and about 40% were production workers (Figure 11). During the pilot survey, it was noted that production workers were hired mainly on a contractual basis, and in a few sectors (such as readymade garments and processed and preserved food), these are primarily women. Of the firms that hired new employees, around 75% also hired women employees. A relatively larger percentage of medium enterprises (85%) hired new women employees following e-commerce integration.

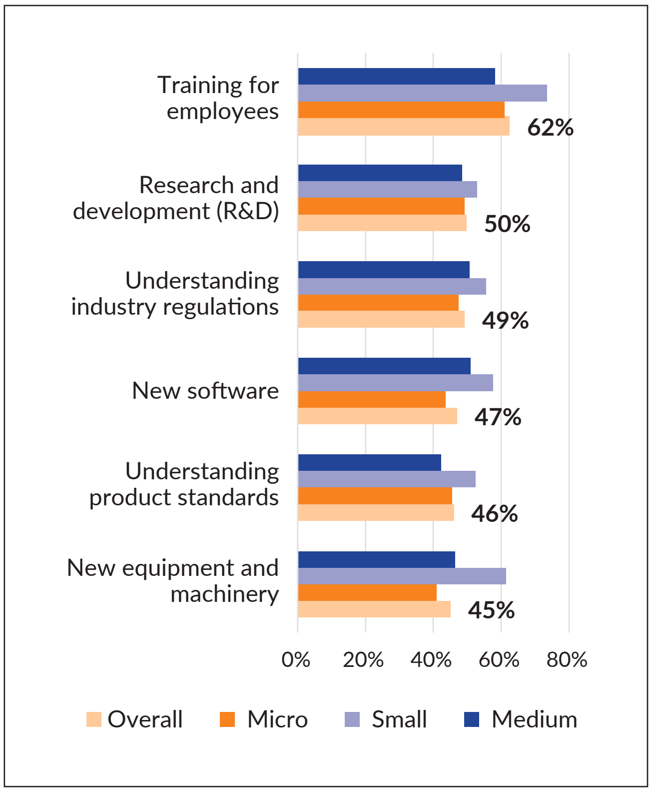
Figure 11: Distribution of enterprises by the roles of the new employees hired after integrating with e-commerce platform (in %)



***Source:** Authors' computation of data from ICRIER's MSME survey*

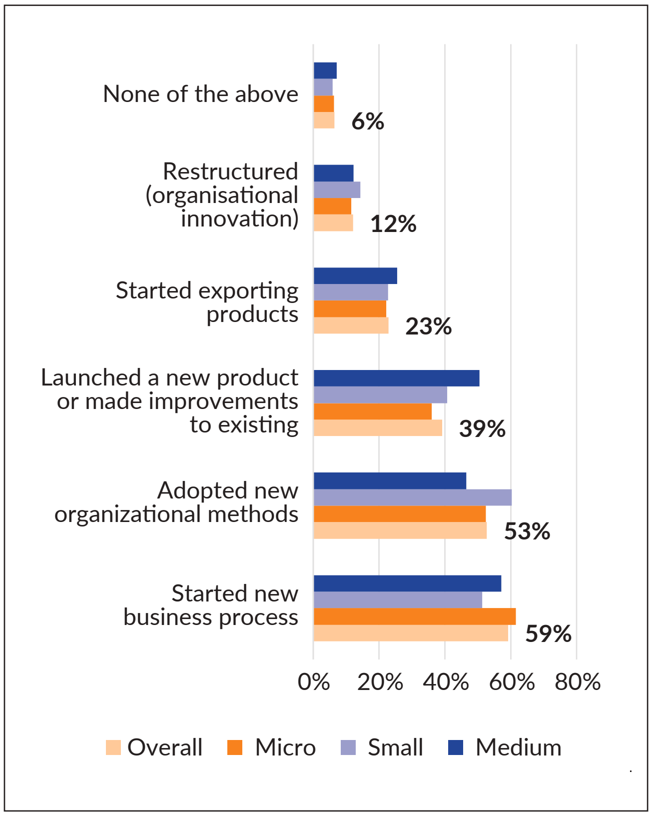
Many enterprises made new investments or introduced innovative practices after integrating on e-commerce platforms. As depicted in Figure 12, most of the firms (62%) invested in employee training and skill development, with a relatively larger percentage of small-sized firms (74%) doing so. As mentioned in the previous paragraph, larger percentage of small-sized firms also hired new employees. It is thus evident that small-sized firms engender greater benefits in terms of employment generation and skill development post-e-commerce integration. This is followed by investments in the R&D of products and in understanding regulations. Over 90% of firms adopted innovative practices after integration (Figure 13). A relatively larger percentage of micro-enterprises introduced new business processes (61%), such as new production or distribution methods or marketing mechanisms; small firms focused on organisational changes such as workplace changes or establishing new business relations (60%); and relatively large percentage of medium firms launched or improved products (50%).

Figure 12: Distribution of enterprises by categories new investments undertaken post integration with e-commerce platforms (in %)



***Source:** Authors' computation of data from ICRIER's MSME survey*

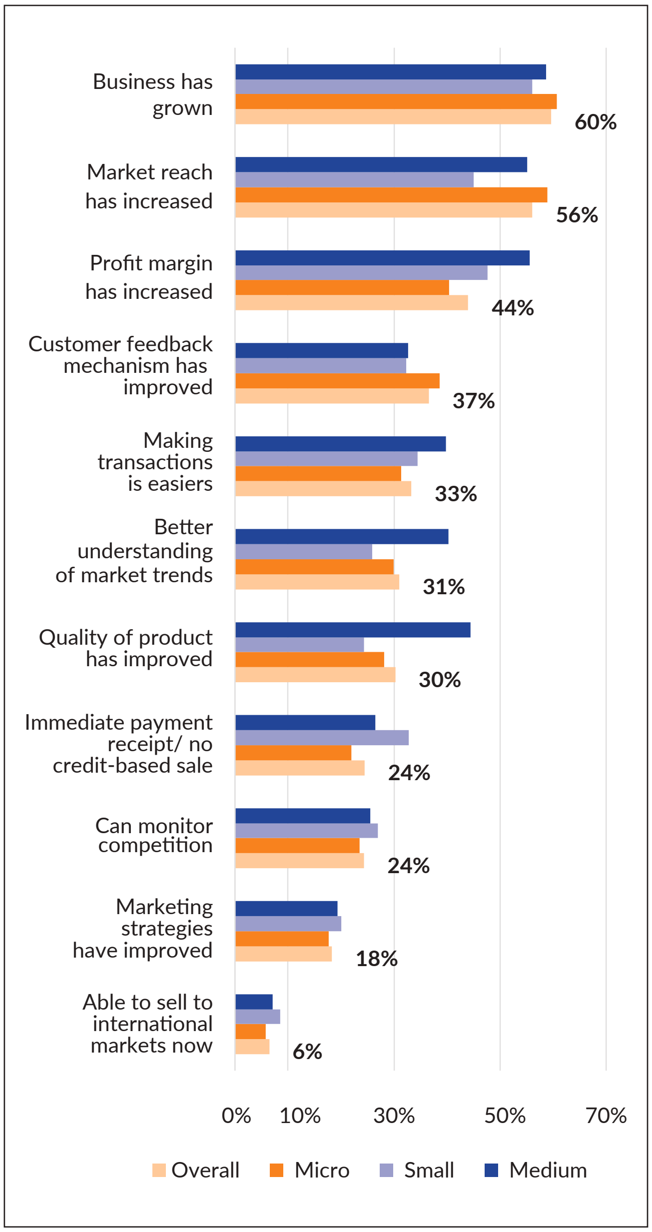
Figure 13: Distribution of integrated enterprises by innovations introduced to sell through e-commerce platforms (in %)



Source: Authors' computation of data from ICRIER's MSME survey

While business growth remains the most important benefit of integrating on e-commerce platforms, medium firms value the qualitative improvement relatively more than micro- and small-sized firms. Around 60% of enterprises cited business growth as the top benefit of e-commerce integration, followed by improved market reach (56%). Micro firms emphasised outreach benefits, including business growth, market reach, and customer feedback, while a relatively larger percentage of medium enterprises highlighted profit margins, market insights, and product quality improvements (Figure 14).

Figure 14: Distribution of enterprises by benefit of e-commerce platform for overall business (in %)

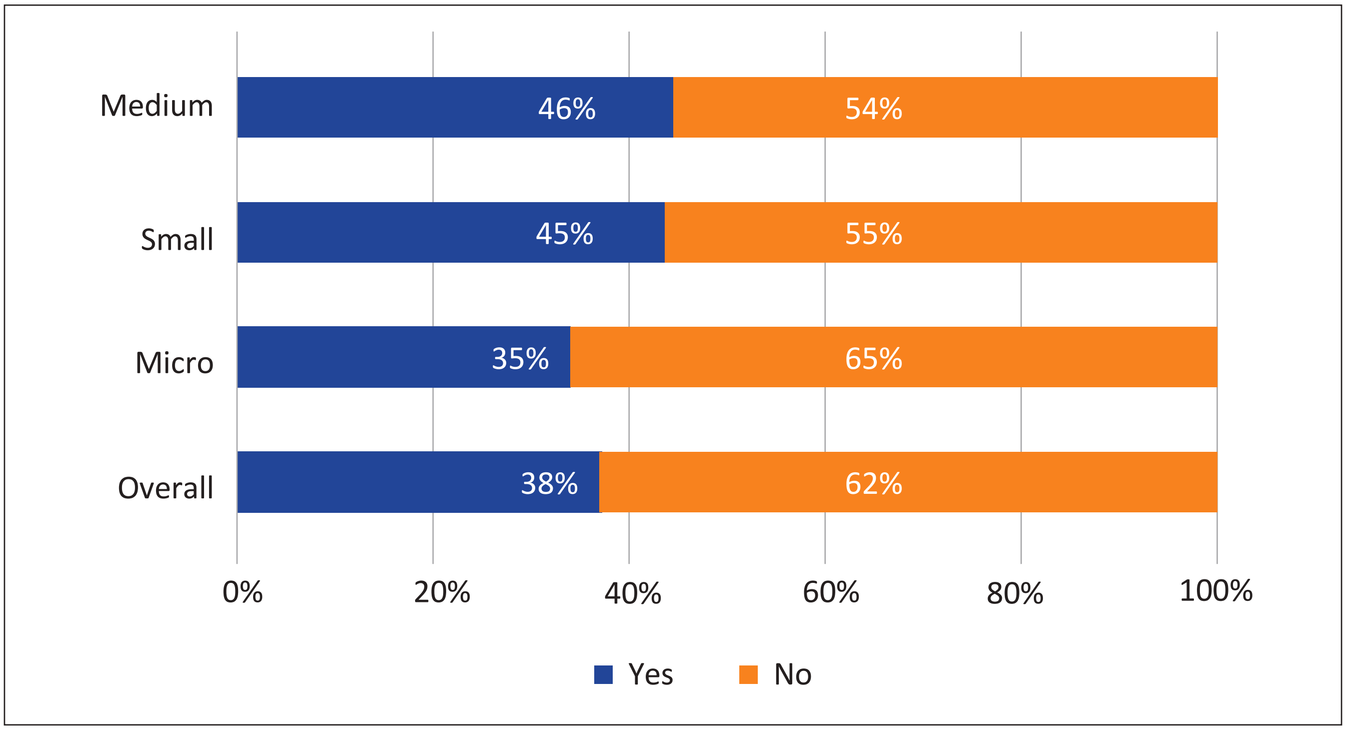


Source: *Authors' computation of data from ICRIER's MSME survey*

Most firms launch deals and pay for sponsored ads to market their products, but a relatively larger percentage of medium enterprises invest in customer experience and deep discounting. Between 33% and 37% of enterprises launch deals, pay for sponsored ads on platforms, and try to maintain high product ratings to market their products. A relatively larger percentage of micro-enterprises use these techniques (36-39%). Comparatively, a larger percentage of medium enterprises invest in prompt customer service (37%), improving shipping performance (31%), and monitoring competition (30%), among others.

A larger percentage of integrated firms also sell through other online channels, thereby improving their online reach. The survey found that compared to non-integrated firms (11%), a much larger share of integrated firms (65%) sells through other online channels, such as their websites and social commerce. This indicates that, more generally, e-commerce-integrated firms have greater digital adoption for revenue generation than non-integrated firms. As depicted in Figure 15, a relatively larger share of medium- (46%) and small-sized firms (45%) are using other online means to sell their products than micro firms (35%). This is primarily because comparatively bigger size firms have greater capacities to manage multiple sales channels and run their website, while a majority of micro firms continue to rely on e-commerce platforms. Greater outreach and higher margins are the most cited reasons for selling through own platforms compared to marketplaces.

Figure 15: Distribution of enterprises by sales through online channels like own website or social commerce other than e-commerce platforms and their size (in %) (all firms)

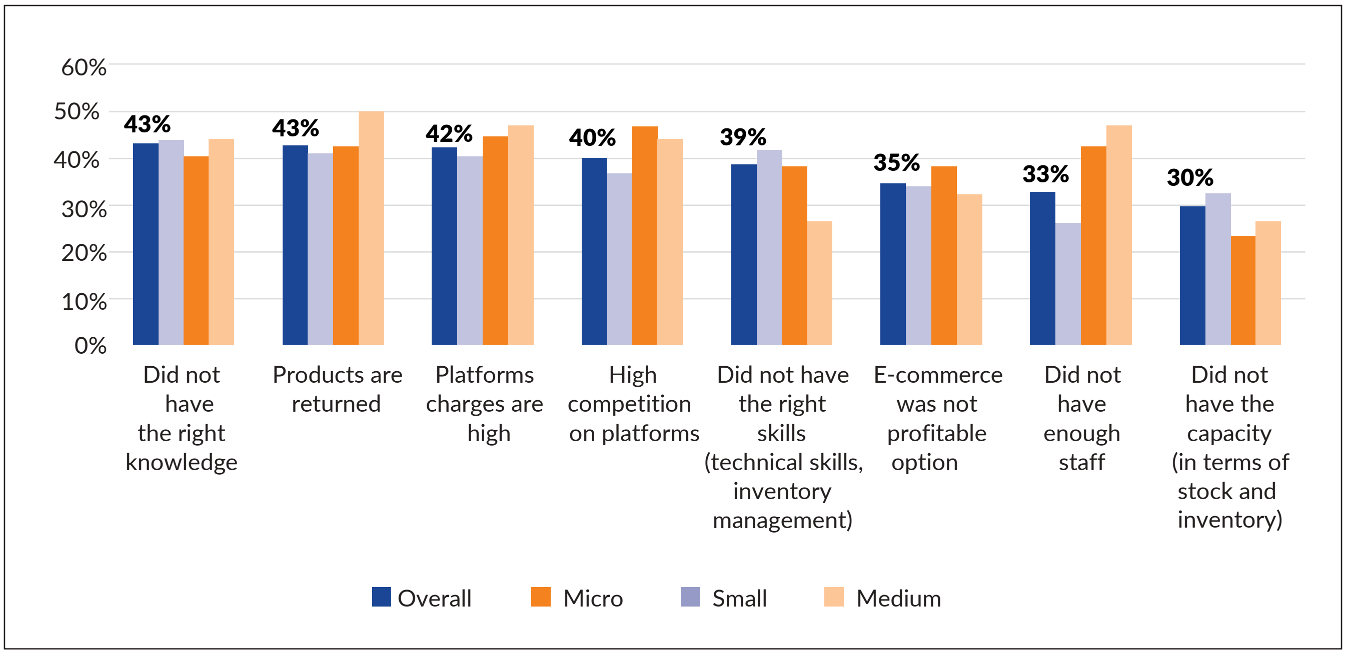


Source: *Authors' computation of data from ICRIER's MSME survey*

Platform charges and compliance burdens are the main barriers to e-commerce adoption, with critical concerns varying by enterprise size. Medium enterprises cite platform charges, skilled staff shortages, and returns as major concerns. Micro-enterprises prefer traditional sales and face strong platform competition, while many small enterprises struggle with knowledge and staffing gaps.

About 1/5th of non-integrated firms joined but quit e-commerce platforms. The primary reason for quitting the platform was lack of knowledge (43% of the respondents who quit platforms). An equal number of firms left these platforms over product returns. A relatively higher percentage of micro-enterprises struggled with skills and capacity constraints, more medium firms faced returns and staffing issues, while a relatively larger percentage of small enterprises cited competition as a prominent reason for quitting e-commerce platforms (Figure 16).

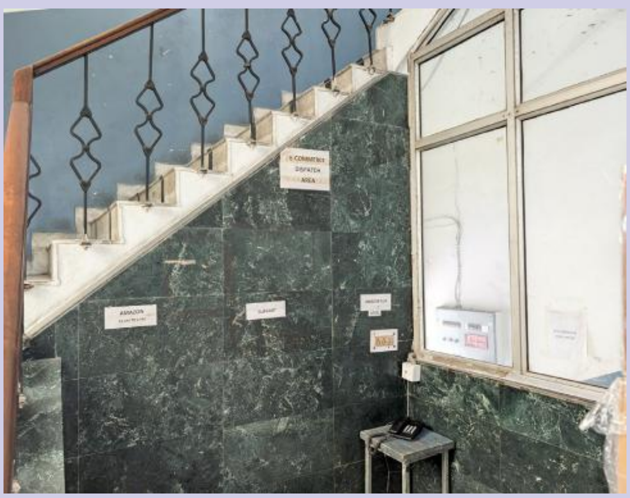
Figure 16: Distribution of enterprises by reason for quitting e-commerce platform (in %)



Source: Authors' computation of data from ICRIER's MSME survey

A significant percentage of firms, most of these being small enterprises, indicated a desire to join e-commerce platforms in the future. About 35% of the surveyed enterprises would like to join e-commerce platforms going forward. The percentage was higher for small-sized enterprises (43%). Better information technology infrastructure and lower platform charges are some factors that can encourage firms to join e-commerce platforms.

Pictures of e-commerce stock holding and dispatch area clicked during pilot survey in July 2024.



MSMEs' access to finance: A challenge that is at last subduable.

Limited access to finance remains one of the most significant challenges for MSMEs, which is a primary obstacle to their scaling potential. Our discussions this year highlighted two major issues: first, securing finance for working capital, particularly small ticket-size, short-duration loans, without which firms struggle to take on multiple orders; and second, high interest rates, largely due to lack of collateral or the burden of multiple ongoing loans. During the pilot interviews, some firms noted that they were being charged interest rates of around 16-17% with collateral, and 24-25% without collateral by commercial banks. At these rates, the struggle is real particularly for export-oriented firms that compete in global markets, operating often on thin profit margins.

One exporting MSME, a contract manufacturer, said that stretched working capital is the biggest hurdle restricting their ability to scale. Although they have the capacity to increase production, their ability to do so is hamstrung by the unaffordability of / inability to secure multiple loans simultaneously. That is because many of their orders tend to be unsupported by formal documentation (as it often follows much later), which could serve as collateral for loans but in the absence of which they end up pledging their machinery instead.

However, digital transformation has led to the emergence of innovative solutions to enhance MSMEs' access to finance, specifically addressing challenges related to collateral and small-ticket loans. Expert discussions highlighted that data-driven digital public infrastructure, such as the Unified Lending Interface and the Account Aggregator Framework, have the potential to revolutionise credit access for underserved segments like MSMEs. Additionally, these frameworks can facilitate quicker disbursement through digital lending. Their effectiveness, however, depends on the depth and quality of available information.

Another promising solution is digital, collateral-free loans offered by non-banking financial corporations (NBFCs) in partnership with e-commerce platforms. These financial intermediaries assess loan eligibility based on sellers' cash-flow data from e-commerce platforms, enabling swift disbursement. Primarily designed as small-ticket loans, they help businesses manage working capital gaps, inventory, and purchase cycles, among other financial needs.

Addressing this challenge requires a collective effort from commercial and public sector banks, NBFCs, microfinance institutions, and private companies. Additionally, regulatory reforms are needed to align MSME credit assessment criteria with the sector's unique characteristics. MSMEs are often denied small-ticket loans from banks due to associated risks and the potential unviability of such loans given the high costs of operating bank branches. Micro-finance institutions and NBFCs are typically better placed to address these financial needs.

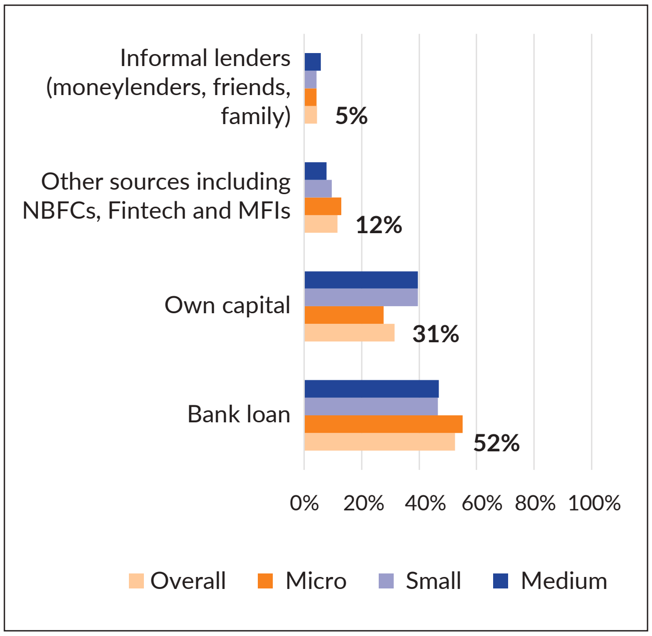
5. MSMEs' Access to Finance

Access to finance remains a challenge for MSMEs but diminishes as firm size grows. Post-COVID-19, about 70% of MSMEs face financial challenges, which include reduced credit availability, difficulties in purchasing inventory, and limited scaling ability. Larger firms report fewer issues, with 44% of small- and 50% of medium-sized firms reporting that they have not faced significant financial challenges, compared to 27% of micro firms.

The nature of financial challenges differs between integrated and non-integrated enterprises. Integrated firms primarily struggle with inventory purchases or working capital to fulfil orders, while non-integrated firms face reduced credit availability and scaling issues. Overall, fewer integrated firms report financial challenges compared to non-integrated ones.

While most firms started operations with their own capital, bank loans are the primary source of finance to continue. More than 50% of the businesses rely on bank loans to run their enterprises, while about a third use their own capital. The use of bank loans is relatively higher among micro-enterprises (55%), compared to 46-47% of small- and medium-sized firms (Figure 17). The previous survey (2023) found that nearly 70% of the surveyed firms used their own capital for starting their business.[35]

Figure 17: Distribution of firms by major source of finance for running the enterprise (in %)

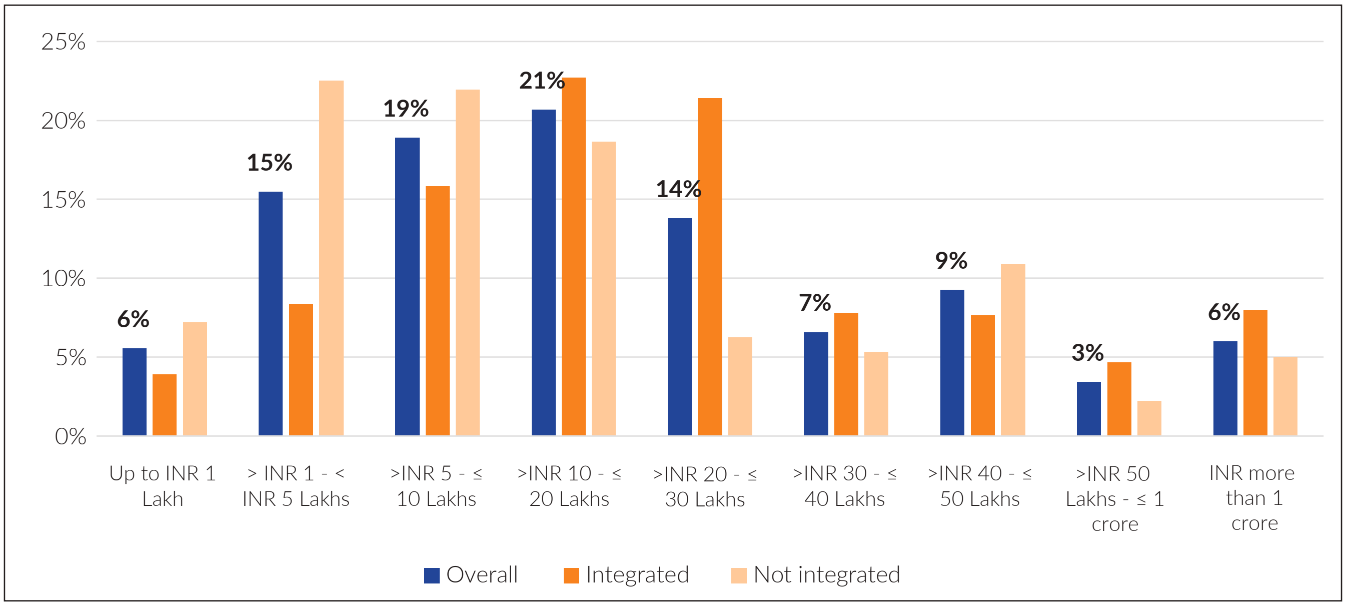


Source: Authors' computation of data from ICRIER's MSME survey

Firms often take small-ticket loans of less than INR 1 lakh to meet their working capital requirements.[36] About 45% of the surveyed enterprises had taken such loans at some point during their lifespan. Of these, approximately 70% were micro-sized. A relatively higher share of integrated firms (nearly 50%) availed small-ticket loans compared to non-integrated firms (40%). Small-sized loans are often essential for survival, providing enterprises with the necessary leverage to fulfil their orders.

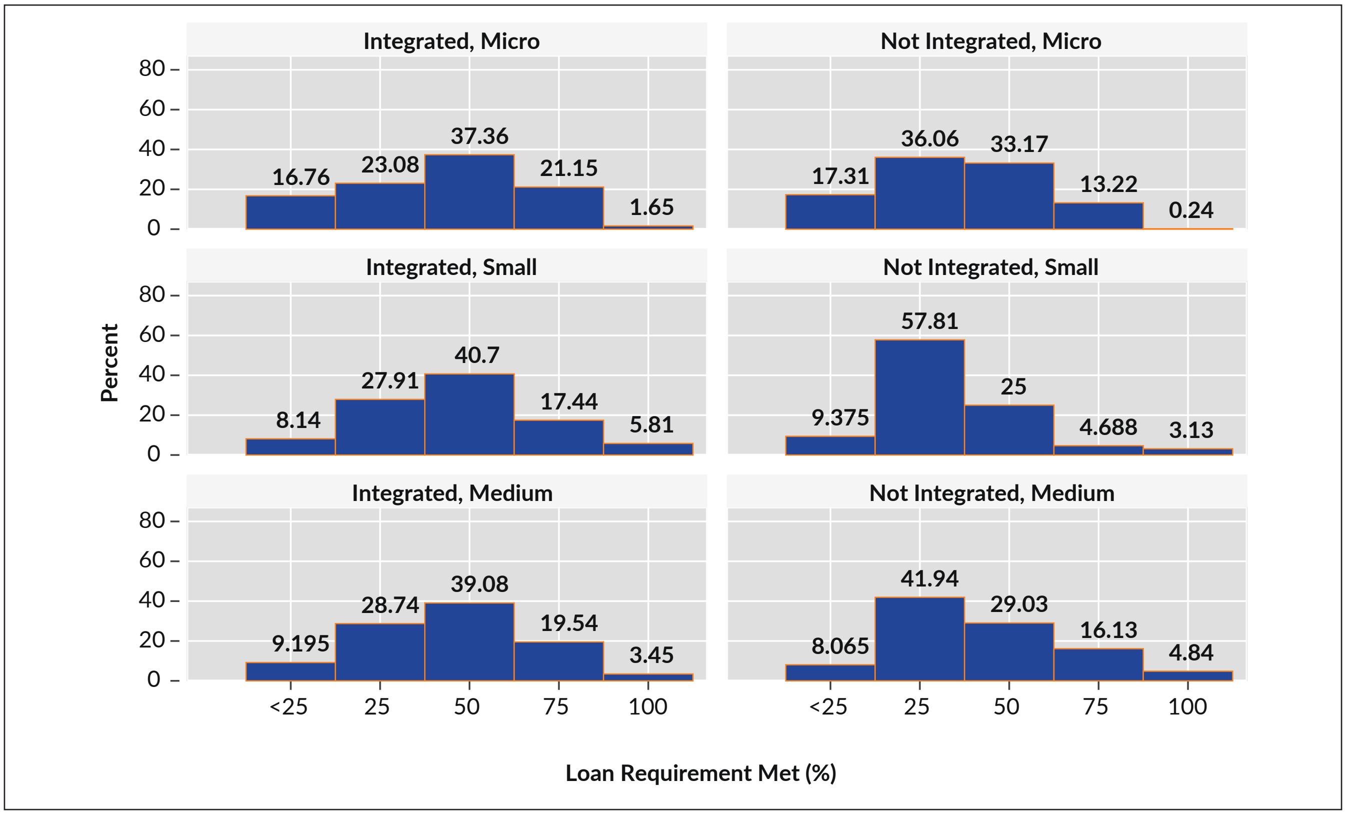
Loans taken do not fully cover the credit needs of the surveyed enterprises. At the time of the survey, 46% of the surveyed enterprises had outstanding loans. Among these, 40% had loans under INR 10 lakhs, less than 10% exceeded INR 1 crore, and the majority (21%) had loans between INR 10 lakhs and INR 20 lakhs (Figure 18). Irrespective of the loan value, more than 98% of enterprises reported that the loan amount was not enough to meet their credit requirement. For 46% of enterprises, the loan covered 25% or less of their credit needs. The credit gap was larger for non-integrated firms and micro-enterprises (Figure 19).

Figure 18: Distribution of enterprises by ticket size (amount) of the current loan (in %)



Source: *Authors' computation of data from ICRIER's MSME survey*

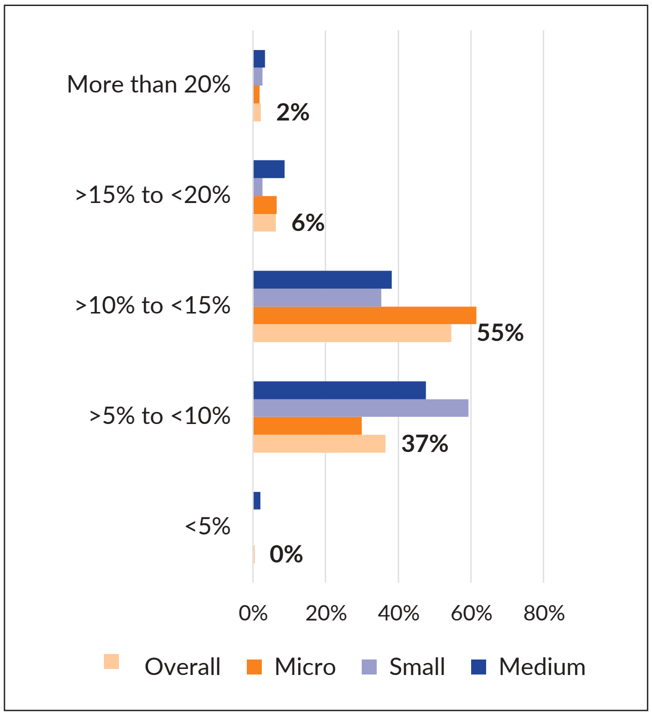
Figure 19: Distribution of enterprises by loan requirement met and e-commerce integration (in %)



Source: *Authors' computation of data from ICRIER's MSME survey*

The cost of credit is considerably high for MSMEs, particularly for micro firms. The average interest rate for MSMEs is about 12.27%, which is much higher than the current repo rate.[37] Micro firms pay an even higher interest rate, averaging around 12.53% (Figure 20). In comparison, the average interest rate for small and medium firms nearly one percent less (at about 11.59%). Interestingly, integrated firms, on average, pay a slightly lower rate of interest (12.15%) than non-integrated firms (12.38%).

Figure 20: Distribution of enterprises by the rate of interest on current loan (in %)

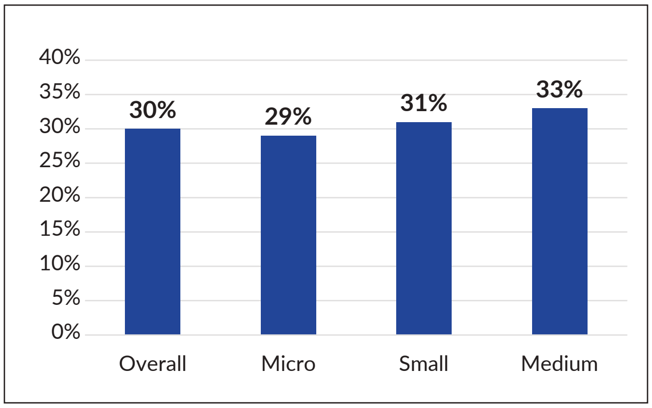


Source: Authors' computation of data from ICRIER's MSME survey

The majority of firms viewed high interest rates as a significant barrier to obtaining loans. About 21% of the surveyed enterprises paid around 15% interest on their loans. Nearly 70% of the surveyed enterprises reported high interest rates as the primary concern while applying for loans, followed by payment terms (51%) and the long waiting period for approval (40%). A relatively higher percentage of non-integrated firms reported facing these difficulties than integrated firms.

MSMEs, especially non-digitalised ones, struggle to secure collateral-free loans. Only 23% of the respondents have ever taken such loans, with over 50% being e-commerce integrated. Among integrated firms, 51% found their e-commerce presence helpful in raising external finance from banks and non-banking financial corporations (NBFCs), particularly medium- (60%) and small-size enterprises (55%) compared to micro firms (48%). Additionally, Figure 21 illustrates that 30% found e-commerce beneficial for collateral-free loans, with medium firms (33%) leading, followed by small (31%) and micro (29%) firms. Thus, along with the digital footprint, firm size plays a key role in accessing formal finance.

Figure 21: Distribution of enterprises reporting that e-commerce presence is useful in raising collateral free loan (in %)



Source: Authors' computation of data from ICRIER's MSME survey

Using Digital Public Infrastructure to Improve MSMEs' Access to Finance: The Case of Account Aggregator Framework

The Account Aggregator (AA) is a data-driven digital public infrastructure that enables secure, consent-based data sharing between financial institutions. This infrastructure is particularly impactful for MSMEs, a segment traditionally constrained by limited access to formal credit due to inadequate financial documentation and collateral-based lending norms. To address these challenges, the AA framework establishes cash-flow-based lending as a viable alternative for MSME lending. Lenders can assess creditworthiness based on actual business cash flows rather than asset-backed guarantees by leveraging real-time transactional data from bank accounts, Goods and Services Tax Network footprints, and other financial records. Additionally, lenders can offer tailored credit products, reduce loan approval times, and lower interest rates for borrowers with healthy cash flow patterns.

Early use-case data shows that the cumulative unsecured business loans disbursed through AAs are estimated at INR 135.77 billion for more than 115 thousand loans. Loans valuing INR 45.25 billion were disbursed to MSMEs in the first half of 2024-25, which translates into INR 7.5 billion monthly disbursements. Unsecured business loans to MSMEs account for around 10% of the total loans facilitated by the AA Framework. Initial numbers indicate the AA ecosystem has significantly impacted MSME loans. Large private banks reported a 30% increase in loans to the Priority Sector Lending category, a 25% reduction in customer acquisition costs, and zero fraud related to bank statements in AA-facilitated MSME lending.

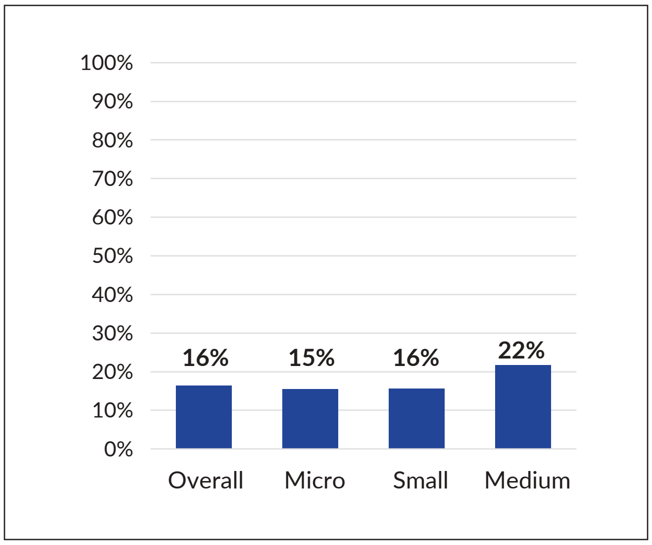
Going forward, there is a need to expand the reach of such systems, particularly to improve micro-enterprises' financial access. New, high frequency robust data sources can be leveraged by partnering with the private sector. For instance, data from e-commerce platforms can be integrated into such networks as the financial information provider. By consolidating cash flow and transaction data on MSMEs, e-commerce platforms have the potential to serve as key digital financial intermediaries. This allows creditors to assess MSMEs' cash flow and risk profiles, enabling the customisation of financial products and pricing. At the same time, public sector banks and non-banking financial corporations can be incentivised to join the network as financial information users.

Source: *Inputs from Sahamati - - Collective of Account Aggregator Ecosystem, ICRIER-ADBI Policy Brief (2024) and primary survey inputs.*

6. Export Orientation of Enterprises

Few MSMEs engage in global trade, with medium enterprises having a higher share of export-oriented firms The survey found 16% of the surveyed enterprises export their products, with medium firms (22%) exporting more than small (16%) and micro (15%) firms, highlighting the role of enterprise size in export potential (Figure 22). Handicrafts, apparel, and processed food products have the highest export share among the surveyed categories.

Figure 22: Distribution of enterprises by the current status of exports (in %)

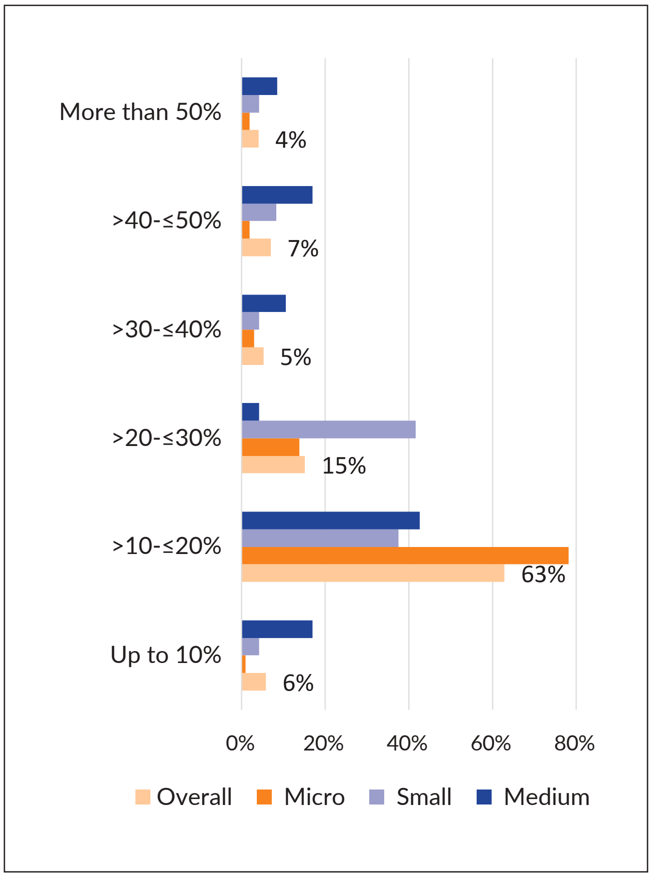


Source: Authors' computation of data from ICRIER's MSME survey

Most firms began exporting post-COVID-19, often after selling in the domestic market at first. About 36% firms started exporting in 2020 and nearly 70% did so after 2014. Only 11% exported in the year they were incorporated, implying that international trade was integral to their business model. Compared to this, about 80% entered international markets two or more years after they were set up, indicating they started tapping export markets for growth.

E-commerce platforms are the preferred export medium for export-oriented integrated firms, including for micro-enterprises. While an equal percentage of integrated and non-integrated firms export, about 93% of the export-oriented integrated firms use e-commerce marketplaces to export. However, the percentage exported through e-commerce platforms is small, with about 85% of the firms exporting less than 30% of their total export sales through e-commerce platforms (Figure 23). This is primarily because at this point, a majority of the e-commerce exports are business-to-consumer in nature, rather than business-to-business.

Figure 23: Distribution of integrated enterprises by the percentage of export sales through e-commerce only (in %)

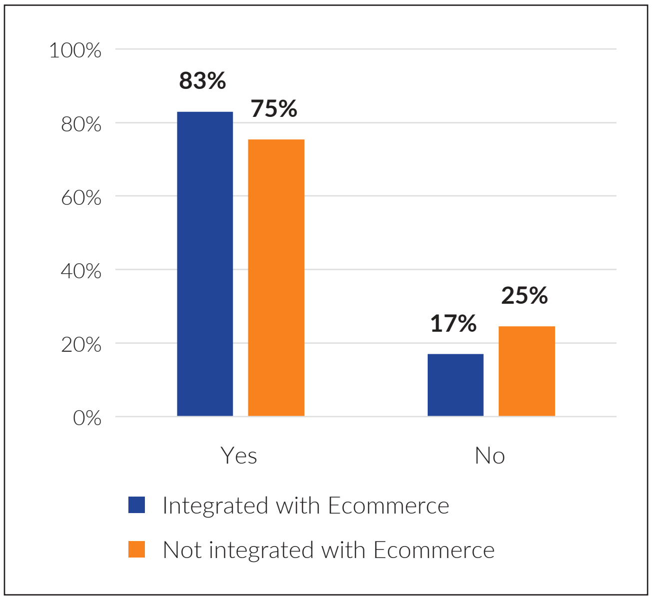


Source: Authors' computation of data from ICRIER's MSME survey

The export potential is not fully tapped yet. Nearly 21% of the firms currently not exporting believe their products have untapped export potential. A higher share of firms in the sports goods (31%) and toys segment

(24%) reported untapped export potential. Integrated firms (24%) recognise this potential more than non-integrated (18%) ones. About 83% of integrated and 75% of non-integrated firms expressed a desire to start exporting within the next 2-3 years (Figure 24).

Figure 24: Distribution of enterprises by the plan to start exporting in next 2-3 years and e-commerce integration (in %)

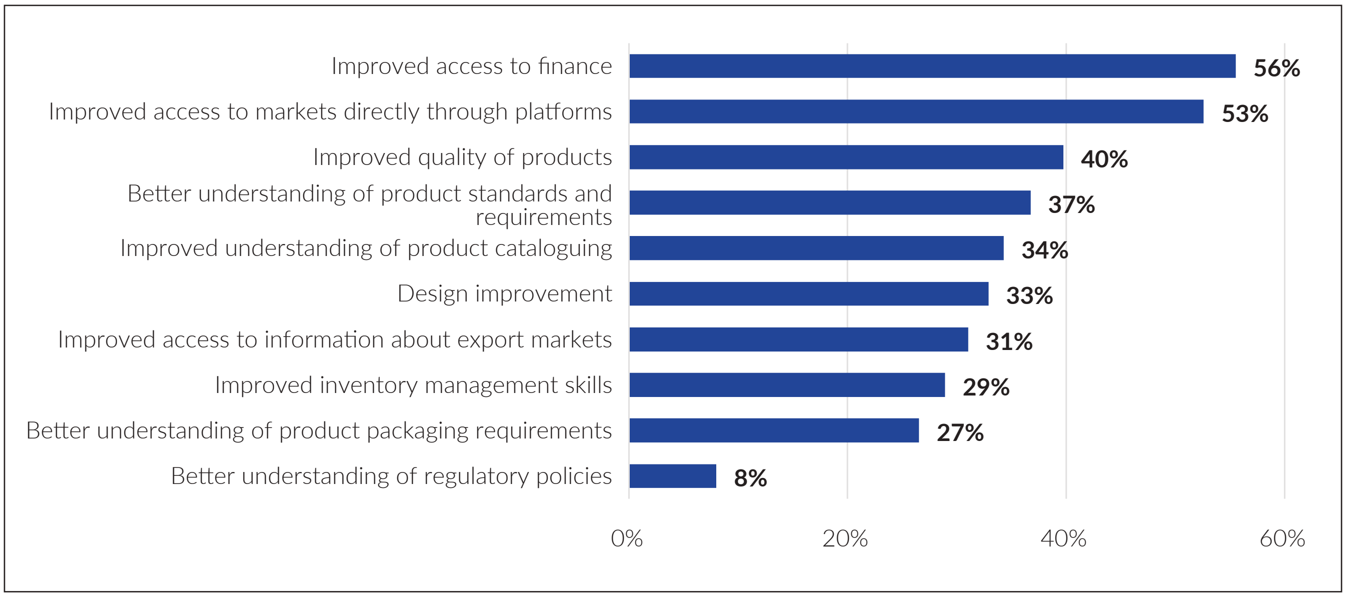


Source: Authors' computation of data from ICRIER's MSME survey

Information gaps and limited access to finance hinder MSMEs' foray into export markets. Nearly 46% of the enterprises reported a lack of information on potential partners and about 42% reported a lack of information on opportunities and requirements in international markets as the most prominent challenges hindering their export potential. Other factors include non-conformity to product standards and a lack of financial resources for export activities.

E-commerce linkages can help in addressing some of these issues. About 38% of the respondents believed that by integrating on e-commerce platforms, they can overcome certain challenges. Of these, about 56% pointed out that e-commerce has the potential to improve access to finance, 53% perceived that they can access markets through e-commerce platforms, and 40% perceived that their presence on e-commerce platforms can help improve product quality to meet international standards (Figure 25).

Figure 25: Distribution of enterprises by perceived benefits of e-commerce integration in building export potential (in %)



Source: Authors' computation of data from ICRIER's MSME survey

7. Challenges and the Way Forward

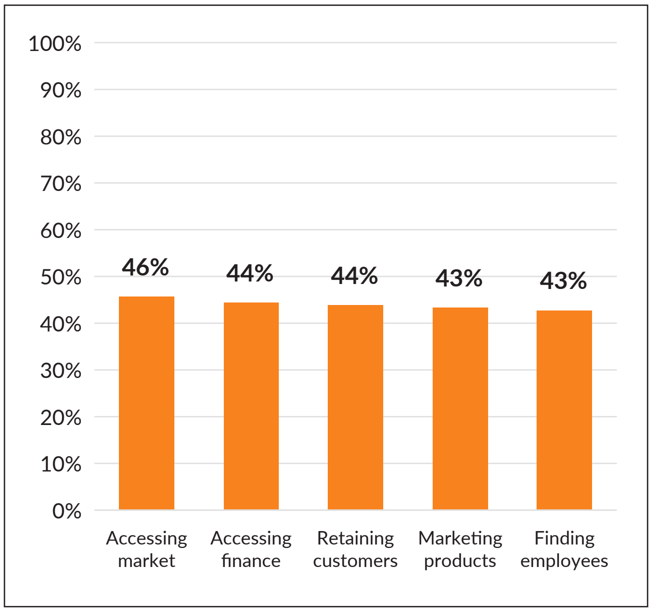
Expanding MSMEs and strengthening the overall ecosystem to enhance their digital adoption is vital to realising their full output and employment-generating potential. Enterprise development focuses on cultivating businesses for growth, scaling, and efficiency. However, various challenges hinder this process, necessitating government and private-sector support to ensure long-term sustainability and growth of MSMEs in India.

Unsurprisingly, access to markets and finance are the most significant challenges confronting non-integrated MSMEs in India. When asked to rank challenges, about one-third of the non-integrated surveyed enterprises listed market access and financing among their top five obstacles to operations and business expansion. Other major challenges included customer retention, hiring the right employees, and effective product marketing (Figure 26 A).

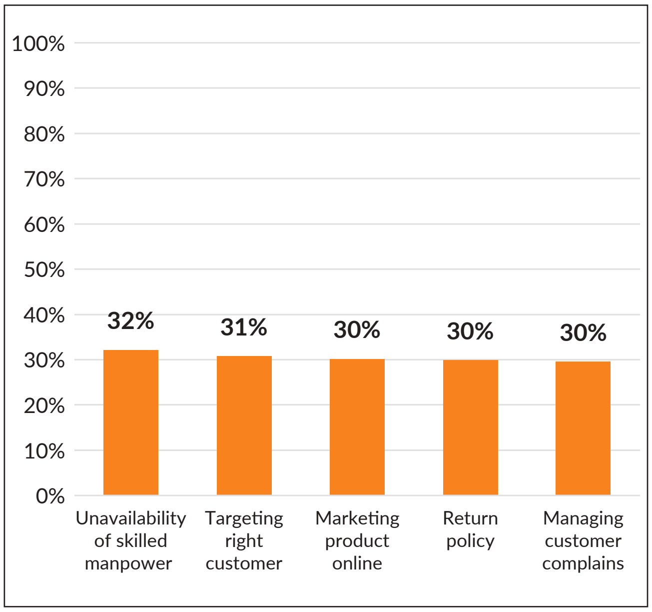
Interestingly, the top challenges of integrated firms differ significantly from those faced by non-integrated firms, with other issues taking precedence over conventional challenges. One-third of the surveyed integrated firms identified a shortage of skilled human resources to handle online business as the top challenge, followed by difficulties in targeting the right customers. Other significant concerns include online product marketing, platform return and refund policies, and handling customer complaints (Figure 26 B). Thus, the core challenges faced by integrated firms primarily involve maximizing the potential of e-commerce platforms. This is not to suggest that other challenges do not exist; however, they are not among the top five concerns by the integrated firms.

Figure 26: Distribution of enterprises by top five challenges faced (in %)

A. No E-commerce Integration

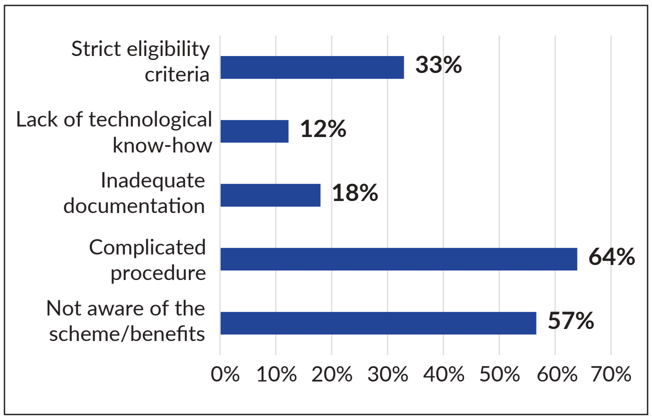


B. E-commerce Integration



While there is no dearth of MSME-focused government schemes that aim to address the conventional issues faced by MSMEs, they are not sufficiently utilised. A significant share of respondents (approximately 55%) reported that they have not accessed these programmes in the past three years. A higher proportion of small- and medium-sized enterprises have not accessed these benefits (55% and 60% respectively; and 54% of micro firms). Among the schemes, the MUDRA loan scheme is the most widely used, followed by the Micro & Small Enterprises Cluster Development Programme. Figure 27 highlights the primary reasons for low uptake, including complex application procedures (64%) and a lack of awareness (57%). Additionally, many MSMEs are unable to qualify due to strict eligibility criteria.

Figure 27: Distribution of overall enterprises by the reason of not availing any schemes/benefits (in %)



Source: Authors' computation of data from ICRIER's MSME survey

The primary challenges faced by integrated and non-integrated firms are notably distinct, necessitating a two-pronged approach to policy intervention. Around 42% of the surveyed MSMEs emphasised the need for government initiatives to improve access to technology, with a higher proportion (47%) among medium-sized enterprises. Additionally, about 40% of respondents recommended strengthening cybersecurity measures, while approximately 39% suggested implementing skill development and training programmes to ensure MSMEs use digital platforms adequately.

Although e-commerce platforms help MSMEs address some traditional challenges, many issues remain unaddressed, necessitating the private sector's intervention. Nearly 48% of the surveyed MSMEs and around half of the micro-enterprises recommend that e-commerce companies resolve the errors and delays on their platforms more effectively. These issues include sudden account de-listing and delays in logistics and payments, among others. The second most common suggestion is to implement stronger measures to prevent fraud, a concern shared by sellers—especially micro firms—just as much as consumers.

Going forward, it is crucial to support firms' digital transformation by enhancing access to technology and upgrading skills and digital literacy of enterprises. ICRIER's survey reveals that e-commerce integration not only supports enterprise development directly through its impact on business growth and hiring, but also through key second-order effects. It is, therefore, important to establish an ecosystem that enables firms to maximise the second-order benefits of e-commerce integration. This can be achieved by developing supportive physical and digital infrastructure that harness the advantages of e-commerce integration.

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Appendix

Table A1: Distribution of the enterprises by their characteristics (in numbers)

Particulars	E-commerce Integration and Size of Enterprises							
	Not Integrated				Integrated			
	Micro	Small	Medium	Overall	Micro	Small	Medium	Overall
OWNER DEMOGRAPHICS								
Age of Primary Owner								
18-24 Years	16	7	3	26	17	6	3	26
25-35 Years	501	92	53	646	514	91	96	701
36-50 Years	262	89	70	421	244	80	78	402
Over 50 Years	38	19	24	81	31	12	19	62
Gender of Primary Owner								
Female	47	8	9	64	58	5	22	85
Male	770	199	141	1,110	748	184	174	1,106
Education of Primary Owner								
School up to 4 Years					1			1
School 5-9 years	7	5	6	18	8	2	4	14
SSC/HSC (completed 10th/12th class)	146	29	27	202	103	20	39	162
Graduate and above	664	173	117	954	693	167	153	1,013
Any other					1			1
FIRM CHARACTERISTICS								
Location								
Agra	56	5	1	62	39	4	1	44
Ahmedabad	28	18	1	47	27	21	4	52
Asansol	43	5		48	35	1	1	37
Bangalore	78	9	12	99	64	15	14	93
Bhubaneshwar	49	3	12	64	44	12	5	61
Chennai	24	4	10	38	25	4	10	39
Coimbatore	4	4	20	28	5	1	36	42
Delhi	31	15	8	54	38	12	10	60

Particulars	E-commerce Integration and Size of Enterprises							
	Not Integrated				Integrated			
	Micro	Small	Medium	Overall	Micro	Small	Medium	Overall
Durgapur	47		1	48	47			47
Guwahati	45			45	40	2	4	46
Hyderabad	51	7	1	59	48	12		60
Jaipur	37	10		47	33	7	3	43
Jalandhar	24	19	20	63	12	4	26	42
Jamnagar	35			35	63		1	64
Kolkata	44	22	8	74	50	28	17	95
Lucknow	20	7	10	37	14	9	7	30
Ludhiana	17	17	18	52	23	14	22	59
Madurai	7	1	5	13	3	2	7	12
Meerut	20	8	3	31	31	3	1	35
Mumbai	77	38	9	124	44	25	12	81
Nagpur	10	6	5	21	17	6	1	24
Pune	21		1	22	20	2	2	24
Rajkot	28			28	47		4	51
Saharanpur	21	9	5	35	37	5	8	50
Exports - No/Yes								
No	671	174	126	971	701	160	145	1,006
Yes	146	33	24	203	105	29	51	185
Year of Establishment								
1960-69	7	2	0	9	17	5	5	27
1970-79	3	1	3	7	18	3	5	26
1980-89	19	6	11	36	17	10	9	36
1990-99	69	37	26	132	86	34	32	152
2000-09	212	56	34	302	197	34	43	274
2010-19	395	86	49	530	379	84	75	538
2020-24	112	19	27	158	92	19	27	138
Number of firms operating out of business cluster and household								
Operating out of business cluster	598	171	150	919	572	159	196	927
Operating out of his/her household	219	36		255	234	30		264

Notes

[1] For this study, the composite criteria of investment in plant and machinery or equipment, and annual turnover of MSME, as prescribed by the Department of MSME, Government of India in 2020, is used as the MSME definition. Micro-enterprises are classified as units with investment in plant and machinery or equipment not more than INR10 million and annual turnover of not more than INR50 million; small enterprises are classified as units with investment in plant and machinery or equipment not more than INR100 million and annual turnover of not more than INR500 million; and medium enterprises are classified as units with investment in plant and machinery or equipment not more than INR500 million and annual turnover of not more than INR2.5 billion. The government proposed a revised MSME definition in the February 2025 Union Budget. However, since the survey was already completed as per the 2020 definition, the same is used for this report and the analysis is based on that definition.

[2] Annual Survey of Unincorporated Sector Enterprises (ASUSE), 2023-24. The survey was conducted during October 2023 to September 2024. National Sample Survey Office, Ministry of Statistics and Programme Implementation, Government of India. The factsheet is accessible here: https://www.mospi.gov.in/sites/default/files/publication_reports/Factsheet_of_ASUSE_2023_24.pdf

[3] The process of registration began in September 2015 with Udyog Aadhar, which was converted to Udyam Registration in 2020 to simplify and streamline the procedures.

[4] Udyam numbers are as of January 2025. Data is accessible at https://udyamregistration.gov.in/Government-India/Ministry-MSME-registration.htm
Please note that ASUSE survey numbers and Udyam numbers are not mutually exclusive. According to ASUSE, about 36.8% of the estimated establishments in the unincorporated non-agricultural sector were found to be registered under different acts/authorities during the survey period.

[5] World Economic Forum, accessible at https://www.weforum.org/stories/2024/06/msme-productivity-gap-global-economy/

[6] World Economic Forum, accessible at https://www.weforum.org/stories/2024/06/msme-productivity-gap-global-economy/

[7] https://www.mckinsey.com/mgi/our-research/a-microscope-on-small-businesses-spotting-opportunities-to-boost-productivity#/

[8] Calculated by from ASUSE 2022-23 and Udyam Portal.

[9] Annual Survey of Unincorporated Sector Enterprises (ASUSE), 2023-24. The survey was conducted during October 2023 to September 2024. National Sample Survey Office, Ministry of Statistics and Programme Implementation, Government of India. The factsheet is accessible here: https://www.mospi.gov.in/sites/default/files/publication_reports/Factsheet_of_ASUSE_2023_24.pdf. This implies that 37% of the enterprises were registered under one Act/ authority, excluding the Companies Act 1956 and 2013.

[10] While the disaggregated numbers of 2023-24 are not out (as of February 2025), in 2022-23, it was about 11% of the total manufacturing sector unincorporated sector enterprises as per ASUSE 2022-23.

[11] University of Stellenbosch Business School (USB-ED, 2019). The definition has been taken from a chapter titled SME Sustainability and Growth in Emerging Market in the Handbook of Research on Sustaining SMEs and Entrepreneurial Innovation in the Post-COVID-19 Era cited as Iwu, C. G. (2021).

[12] See International Labour Organization's note on enterprise development, accessible at https://www.ilo.org/resource/enterprise-development

[13] Calculated by from ASUSE 2022-23 and Udyam Portal.

[14] Other challenges include the issue of informality, and regulatory and infrastructural bottlenecks.

[15] International Trade Centre (ITC) (2019). Big Money for Small Businesses: Financing the Sustainable Development Goals, SME Competitiveness Outlook, 2019, Geneva: ITC. Accessible at https://intracen.org/media/file/2523

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[22] For details see https://tfig.unece.org/contents/market-access-challenges-SMEs.html (accessed on 12 September 2022).

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[29] ICRIER (2023). Annual Survey of MSMEs.

[30] ASUSE (2023-24), (2022-23).

[31] ASUSE (2022-23). The list omits pharmaceutical manufacturing, which has the highest concentration because they are not sold on general e-commerce platforms.

[32] Singh & Hess (2017) and Porfírio et al. (2021)

[33] During the survey, enterprises were provided with turnover ranges and asked to choose the most appropriate one. To determine the average value, the midpoint of each selected range was used for calculation. While this is an estimate, it enables us to assess the magnitude of change effectively.

[34] ASUSE (2023-24).

[35] ICRIER (2023).

[36] A "small ticket" loan lacks a standard definition. While loans under INR 1 lakh are considered small for individuals, the range varies for firms. During the survey enterprises were asked if they have taken a loan of less than INR 1 lakhs.

[37] The repo rate at the time of the survey was 6.5% and has been at this level since February 2023. While it may not be completely accurate to take a simple average of the interest rates reported by firms as lending rates are a function of several factors, yet most of the factors culminate from the nature of the enterprise and therefore, a simple average is used to bring out the sector average.

Worksheet

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ISBN 978-81-986536-3-5

9 788198 653635